

Persistent activation of TRPM4 triggers necrotic cell death characterized by sodium overload

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Sodium influx and overload are frequently observed in human tissue injuries. Whether sodium overload imposes a causative effect on necrotic cell death and the mechanism involved are unclear. Here we identify necrocid 1 (NC1) as a compound that induces necrotic cell death through sodium overload, termed NECSO for necrosis by sodium overload. NC1 targets the transient receptor potential cation channel subfamily M member 4 (TRPM4), a nonselective monovalent cation channel, to promote Na⁺ influx and necrosis. TRPM4-deficient cells are resistant to NC1-induced NECSO. NC1 specifically activates human TRPM4, not mouse TRPM4, because of differences in a transmembrane region, as revealed by domain swapping and molecular docking. Gain-of-function mutations in human *TRPM4* linked to cardiac arrhythmias show increased vulnerability to NECSO triggered by NC1 or 2-deoxy-D-glucose. Chemical screening identified NECSO inhibitors that block necrosis induced by NC1 or energy depletion. These findings provide insights into regulated Na⁺ influx-mediated necrosis and its implications for disease.

Excessive sodium contributes to severe human diseases such as hypertension, cardiovascular and kidney diseases^{1,2}, brain edema³ and diabetes⁴. Dysregulated sodium influx disrupts osmotic balance, causing tissue edema⁵ linked to conditions such as tissue injury⁶, stroke⁷ and tumors⁸. Whether sodium overload directly triggers cell death, as seen with calcium, iron or copper^{9–11}, remains unclear.

Sodium concentration is tightly regulated by Na⁺–K⁺ adenosine triphosphatase (ATPase) and ion channels, maintaining low intracellular and high extracellular levels¹². Sodium influx is essential for membrane potential, action potential propagation and cell communication, regulated by various factors such as ligands and signaling pathways^{13,14}.

We previously identified necrocid 1 (NC1) as a potent inducer of regulated necrosis distinct from known cell death pathways¹⁵. Here, we found that NC1 triggers massive sodium influx, rapid membrane depolarization and cell edema. CRISPR–Cas9 screening revealed

transient receptor potential cation channel subfamily M member 4 (TRPM4), a nonselective cation channel, as NC1's target, inducing necrosis through sodium overload. The manipulation of necrosis by sodium overload (NECSO) may benefit in protecting cells or tissues from energy-depletion-induced injuries.

Results

TRPM4 is essential to NC1-induced necrotic cell death

We previously described a lethal chiral compound, NC1 (Extended Data Fig. 1a), which induces robust immunogenic necrotic cell death in specific cancer cell lines through unknown mechanisms¹⁵ (Extended Data Fig. 1a–c and Supplementary Video 1). Attempts to suppress NC1-induced necrosis using inhibitors of apoptosis (Z-VAD), necroptosis (Nec-1), autophagy (CQ and 3-MA) or ferroptosis (Fer-1, Liprox-1 and DFO) were unsuccessful (Extended Data Fig. 1d). Consistent with

previous observations¹⁵, these data support that NC1 induces a distinct cell death pathway.

To uncover its molecular mechanism, we conducted a genome-wide CRISPR–Cas9 screen in NC1-sensitive MCF7 cells stably expressing Cas9 (Supplementary Fig. 1a,b). High-quality single guide RNA (sgRNA) transfection was confirmed through deep sequencing, with 99.9% of sgRNAs successfully transfected (Supplementary Fig. 1c). Screening revealed key genes for NC1 resistance in the NC1-resistant cell pool (Supplementary Fig. 1d), analyzed using the MAGeCKFlute pipeline¹⁶. TRPM4 emerged as the top candidate, consistently showing extraordinary enrichment scores (Fig. 1a).

To validate the effect of TRPM4, TRPM4-knockout (KO) MCF7 cells were generated using three gRNAs. TRPM4 deletion completely blocked NC1-induced cell death, as shown by cell viability and lactate dehydrogenase (LDH) release assays (Fig. 1b,c and Supplementary Fig. 2a). Reintroducing human TRPM4 (hTRPM4) restored NC1 sensitivity, while the TRPM4 inhibitor 9-phenanthrol¹⁷ dose-dependently alleviated NC1-induced cell death (Fig. 1d and Supplementary Fig. 2b,c). Furthermore, the ion-impermeable TRPM4 mutant D984A (ref. 18) failed to rescue the necrosis phenotype in TRPM4-KO cells, confirming that TRPM4's ion channel activity is required for NC1-induced necrosis (Fig. 1d). These results establish that TRPM4 is essential for NC1-induced cell death.

NC1 directly binds to and activates TRPM4

We hypothesized that TRPM4 is the direct target of NC1, given its top ranking in our CRISPR screen. We firstly performed cellular thermal shift assay (CETSA) to analyze the interaction between NC1 and TRPM4 in HEK293T cells stably expressing TRPM4. NC1 stabilized TRPM4 at elevated temperatures, whereas DMSO or its inactive stereoisomer (NC1i) did not (Fig. 1e and Supplementary Fig. 2d). Surface plasmon resonance (SPR) analysis confirmed that NC1 directly binds to TRPM4 with an affinity of 0.88 μ M (Fig. 1f and Supplementary Fig. 2e,f). These results suggest that NC1 specifically interacts with TRPM4 to induce cell death.

To investigate NC1's effect on TRPM4, whole-cell patch-clamp recordings were conducted in HEK293T cells stably expressing TRPM4–EGFP. NC1 robustly induced TRPM4-dependent currents (half-maximal effective concentration (EC_{50}) $\approx$ 306.3 nM), which were inhibited by the TRPM4 inhibitor 9-phenanthrol, confirming TRPM4 specificity (Extended Data Fig. 2a and Supplementary Fig. 2g,h). In wild-type (WT) MCF7 cells, NC1 elicited strong currents while TRPM4 KO blocked this effect (Fig. 1g,h). Additionally, NC1i failed to activate TRPM4 (Extended Data Fig. 2b,c). In a cell-free system, TRPM4 was gated upon NC1 exposure on the outside-out membrane patch (rather than the inside-out patch) (Extended Data Fig. 2d–g). Remarkably, NC1 triggered a sustained and increased current along with the time course and the current did not diminish even after washout, suggesting the continuous and irreversible activation of TRPM4 upon NC1 exposure (Fig. 1i and Extended Data Fig. 2h). These results indicate that NC1 directly binds to TRPM4 and triggers a persistent activation of TRPM4.

NC1-induced necrosis through TRPM4 is driven by sodium influx

TRPM4 is a calcium-activated, nonselective monovalent cation channel protein^{19,20}. As most cations *in vivo* are metals, we analyzed major intracellular metal levels using inductively coupled plasma mass spectrometry (ICP-MS), including Na, K, Ca, Mg and Fe. Upon NC1 treatment, sodium levels increased while potassium decreased, whereas Ca, Mg and Fe levels remained stable, suggesting robust sodium influx and potassium efflux (Fig. 2a). Ion-sensitive probes confirmed these changes; CoroNa green fluorescence (sodium) increased and PBFI fluorescence (potassium) decreased in NC1-treated MCF7 cells (Fig. 2b,c). The above results indicate that sodium influx and potassium efflux occur with NC1-induced necrosis.

To determine whether sodium influx or potassium efflux drives NC1-induced necrosis, we assessed its killing activity in isotonic extracellular buffer solution (EBS) devoid of serum and amino acids. The treatment was limited to 6 h to elicit NC1's effect while minimizing starvation-induced cell death. NC1 exhibited similar cytotoxicity in EBS compared to normal culture medium and calcium depletion from EBS did not affect cell death (Fig. 2d,e). To investigate sodium's role, NaCl in the medium was replaced with *N*-methyl-D-glucamine (NMDG)-Cl, a nonpermeable cation that maintains osmolarity. Sodium depletion significantly blocked NC1-induced cell death, indicating sodium influx as a critical factor (Fig. 2d,e). Conversely, raising extracellular potassium concentration from 5 mM to 25 mM to inhibit potassium efflux did not affect cell death, suggesting that potassium efflux is a concomitant not causative event (Fig. 2d,e). Intracellular concentrations of sodium and potassium were confirmed, with ICP-MS indicating the blockage effect of sodium influx and potassium efflux under the above conditions (Fig. 2f). Notably, Na⁺ depletion with NMDG-Cl blocked sodium influx and cell death without perturbation of potassium efflux (Fig. 2e,f). Potassium channel inhibitors targeting NKCC (Na⁺–K⁺–2Cl[–] cotransporter) and KATP (ATP-sensitive potassium channel) also had no impact on cell death (Extended Data Fig. 3a). To confirm the dependency on sodium influx, reducing sodium concentrations in EBS progressively decreased NC1's killing activity (Fig. 2g). Substitution of sodium with other monovalent cations inhibited cell death to varying degrees (Fig. 2h), highlighting sodium influx as the primary factor. TRPM4 KO in NC1-treated MCF7 cells abolished sodium influx, confirming TRPM4's role in mediating this effect (Fig. 2i). Collectively, these data suggest that NC1 triggers TRPM4-dependent necrotic cell death through the induction of sodium influx; therefore, we refer to this specific type of necrosis as necrosis by sodium overload (NECSO).

TRPM4 is a Ca²⁺-activated cation channel mediating cell membrane depolarization²⁰. We hypothesized that NC1-induced sodium influx causes depolarization. Using a voltage clamp (Extended Data Fig. 3b,c) and the membrane potential probe DiBAC₄(3) (Extended Data Fig. 3d,e), we observed membrane depolarization within 5–10 min of NC1 exposure. Because of electroneutrality, sodium influx is typically accompanied by chloride influx. Chloride levels, measured with the chloride quencher MQAE, increased upon NC1 treatment (Extended Data Fig. 3f,g). Additionally, inhibitors targeting anion channels or transporters effectively blocked NECSO to varying degrees (Extended Data Fig. 3h). Sodium overload also increases intracellular osmolarity, leading to water influx, cell swelling and edema. Transmission electron microscopy (TEM) revealed severe swelling of organelles, including the mitochondria, endoplasmic reticulum and Golgi, as well as membrane rupture, in NC1-treated cells (Extended Data Fig. 3i). Treatment with D-mannitol, a nonpermeant oncotic agent, increased extracellular osmolarity and delayed NECSO progression (Extended Data Fig. 3j). These findings indicate that membrane depolarization and osmotic imbalance-induced cell swelling are defining features of NECSO.

NC1 specifically targets hTRPM4

We investigated whether TRPM4 expression correlates with cellular sensitivity to NECSO. Among nine human cancer cell lines, MCF7, MB468, HCC1143 and PC3, which highly expressed TRPM4, were strongly sensitive to NC1. HEK293T, U-2OS and A549 cells, with moderate TRPM4 expression, showed intermediate sensitivity, while SW620 and HeLa cells, with negligible TRPM4 expression, were least sensitive (Fig. 3a,b and Extended Data Fig. 1b). Notably, mouse and hamster cells, including normal and cancer lines, were over 1,000-fold more resistant to NC1 than human cells, despite some mouse cells expressing TRPM4 at levels similar to several human cell lines (Fig. 3a and Extended Data Fig. 4a). This suggests species-specific differences in TRPM4 confer NC1 resistance.

To test whether NC1 specifically targets hTRPM4, we re-expressed hTRPM4 or mouse TRPM4 (mTRPM4) in TRPM4-KO MCF7 cells. Only

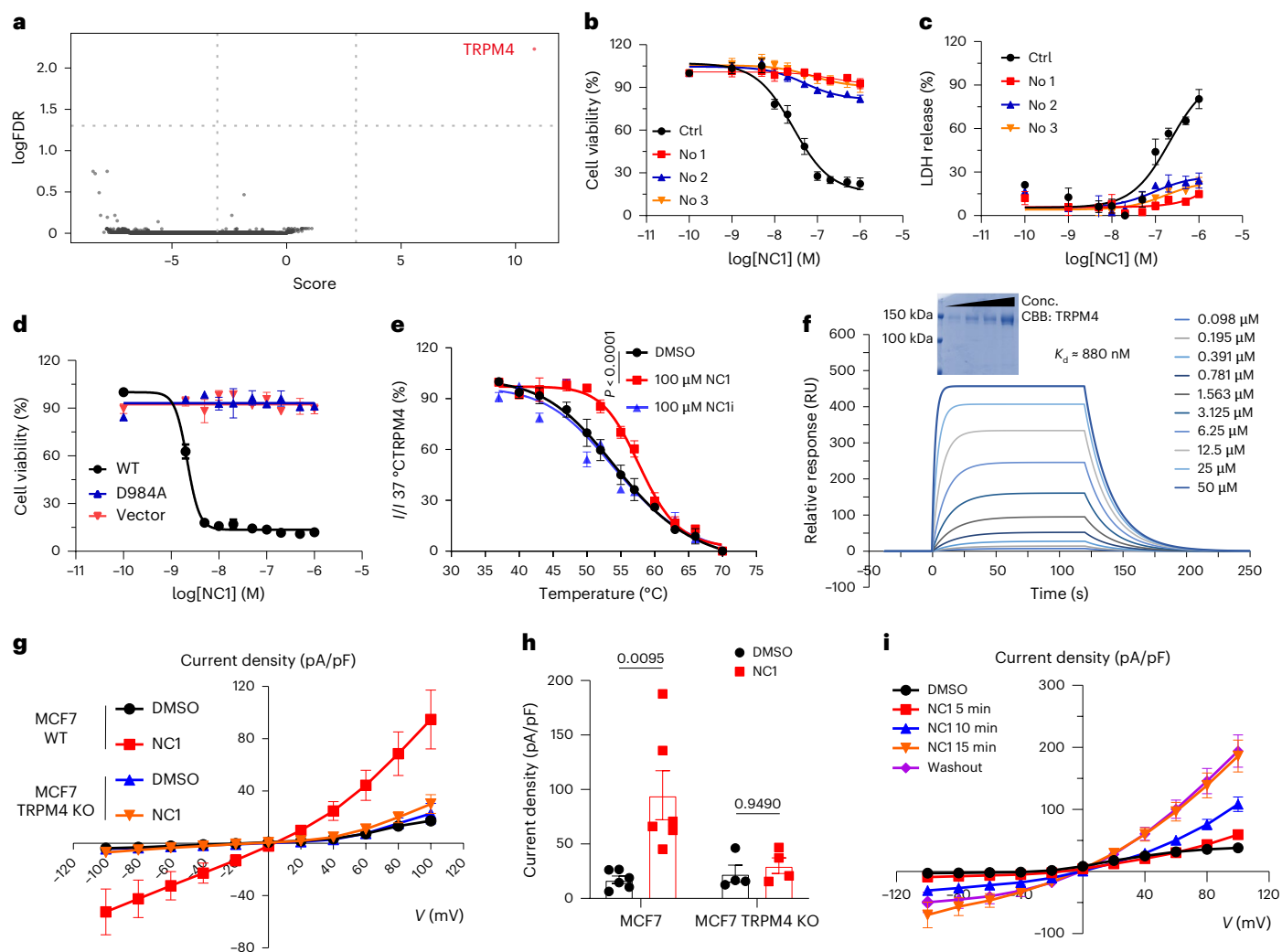


Fig. 1 | NC1 triggers a distinct cell death pathway dependent on TRPM4.

a, Beta scores and false discovery rate (FDR) of sgRNA-targeted genes analyzed by the MAGeCKFlute-RRA of NC1 treatment and control in the mentioned CRISPR–Cas9 screening. The horizontal dotted line on the y axis corresponds to a value of 1.30 ($-\log_{10}(0.05)$). The cutoff for the x axis is set at ± 3.03 . **b**, Dose-dependent cytotoxicity of NC1 was assessed 24 h after NC1 treatment using the CTG assay in MCF7 cells (control and TRPM4 KO with three different sgRNAs). **c**, Cell death assay measured by LDH release in MCF7 TRPM4-KO cells treated with NC1 for 24 h. **d**, Dose-dependent cytotoxicity of NC1 in MCF7 TRPM4-KO cells with re-expression of WT TRPM4 or D984A mutant. **e**, CETSAs of NC1 or NC1i in HEK293T cells stably expressing TRPM4–EGFP. The graph represents the relative TRPM4 band intensities of the indicated samples as the mean $\pm$ s.d. from three independent experiments. **f**, Purification with size exclusion chromatography of hTRPM4 (inset). Purified protein was loaded in dose-dependent concentrations

(Conc.) and stained with Coomassie brilliant blue (CBB). NC1 binds to the hTRPM4 protein as shown by SPR measurements. **g**, The whole-cell current density–voltage relation of MCF7 and MCF7 TRPM4-KO cells activated by NC1. **h**, Statistics of hTRPM4 current density at +100 mV in **g**. **i**, The whole-cell current density–voltage relation of NC1 activating hTRPM4 stably expressed in HEK293T cells with duration recordings. Viability data represent the mean $\pm$ s.d. of $n = 3$ replicates from one representative of three independent experiments. Electrophysiological data represent the mean $\pm$ s.e.m. of ($n = 6$ for MCF7 cells and 4 for MCF7 TRPM4-KO cells in **g**; $n = 7$ for vehicle, 10 for NC1 5 and 10 min, 7 for NC1 15 min and 5 for NC1 washout in **i**) independent cells. Immunoblot data show one representative of three replicates from three independent experiments. Statistical analysis was performed using a one-way ANOVA (GraphPad Prism 9.0.0) in **e** and two-way ANOVA in **h**.

hTRPM4 restored NC1 sensitivity (Fig. 3c). CETSAs confirmed that NC1 binds hTRPM4 but not mTRPM4 (Fig. 3d). Additionally, NC1 activated hTRPM4 channel activity but failed to activate mTRPM4 (Fig. 3e), demonstrating NC1's specificity for hTRPM4.

We hypothesized that TRPM4 sequence differences among species account for their distinct sensitivities to NC1. To test this, we used HeLa cells, which lack detectable TRPM4 and are resistant to NC1. Transient expression of hTRPM4 restored NECSO, characterized by ‘balloon-like’ necrotic cells, upon NC1 treatment (Fig. 3f). This system allowed us to evaluate the role of TRPM4 variants in NECSO.

TRPM4 homologs from mouse and other species, unlike hTRPM4, showed no sensitivity to NC1-induced NECSO (Fig. 3g and Extended Data Fig. 4b–d). Additionally, other TRPM family members, including

the closely related TRPM5, did not respond to NC1 (Fig. 3h and Supplementary Fig. 3a–d). These findings demonstrate that NC1 is a highly specific agonist for hTRPM4, enabling NECSO exclusively in cells expressing hTRPM4.

TRPM4 TM region determines NC1 response

TRP proteins from different species exhibit varied sensitivity to stimuli, aiding in identifying receptor domains involved in ligand binding or activation^{21,22}. hTRPM4 and mTRPM4 share 88.4% sequence similarity and similar tetrameric structures, with conserved regulatory modifications such as phosphorylation, sumoylation and glycosylation^{23–26}. However, their NC1 sensitivity differs sharply (Fig. 3c). We performed domain-swapping analysis to identify the structural domains required

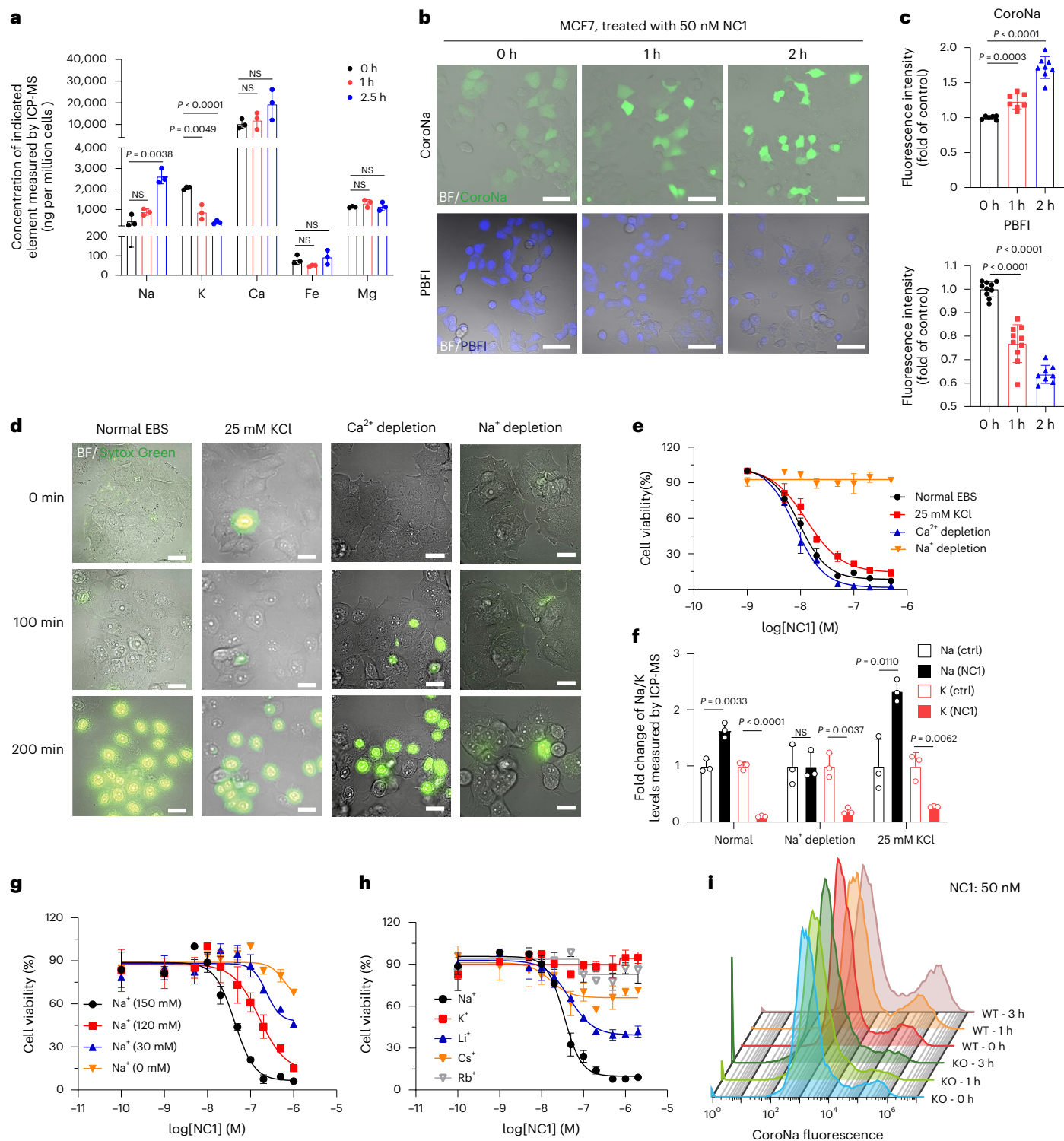


Fig. 2 | Sodium influx is an essential event for NC1-induced cell death.

a, Sodium, potassium, calcium, iron and magnesium levels were assessed by ICP-MS in MCF7 cells treated with 50 nM NC1 for 0 (control cells), 1 and 2.5 h. The data are represented as the mean $\pm$ s.d. of triplicate samples. **b**, MCF7 cells were loaded with sodium indicator CoroNa (green) and potassium indicator PBFI (blue) and then treated with 50 nM NC1 for 0, 1 or 2 h. Scale bars, 50 μ m. **c**, Statistics of fold change of CoroNa or PBFI fluorescence intensity. Data shown are the mean $\pm$ s.d. Each dot represents data from an individual image in **b**. **d**, Representative time-lapse images of MCF7 cells treated with 50 nM NC1 in different EBSs: normal, increased concentration of KCl to 25 mM, calcium-free and sodium-free (osmotic pressure was maintained by an equal concentration of NMDG-Cl). Scale bars, 25 μ m. **e**, The cell viability of MCF7 cells treated with NC1 for 6 h in different EBSs described in **d**. **f**, Sodium and potassium levels were assessed by ICP-MS in MCF7

cells treated with 50 nM NC1 for 3 h in different EBSs (normal, sodium-free and 25 mM KCl). Results are plotted as the fold change over control. The data are represented as the mean $\pm$ s.d. of triplicate samples. **g**, Cell viability of MCF7 cells treated with NC1 for 6 h in EBSs containing varying concentrations of Na^{+} (osmotic pressure was maintained by an equal concentration of NMDG $^{+}$). **h**, The cell viability of MCF7 cells treated with NC1 for 6 h in EBSs containing different monovalent cations (Na^{+} in EBS was replaced by an equal concentration of K^{+} , Li^{+} , Cs^{+} or Rb^{+}). **i**, Intracellular sodium levels measured by CoroNa in MCF7 WT and TRPM4-KO cells treated with 50 nM NC1 for 0, 1 and 3 h. Viability data represent the mean $\pm$ s.d. of $n = 3$ replicates from one representative of three independent experiments. Micrograph and flow cytometry data show one representative of at least three replicates from three independent experiments. Statistical analysis was performed using a one-way ANOVA (GraphPad Prism 9.0.0) in **a**, **c**, **f**.

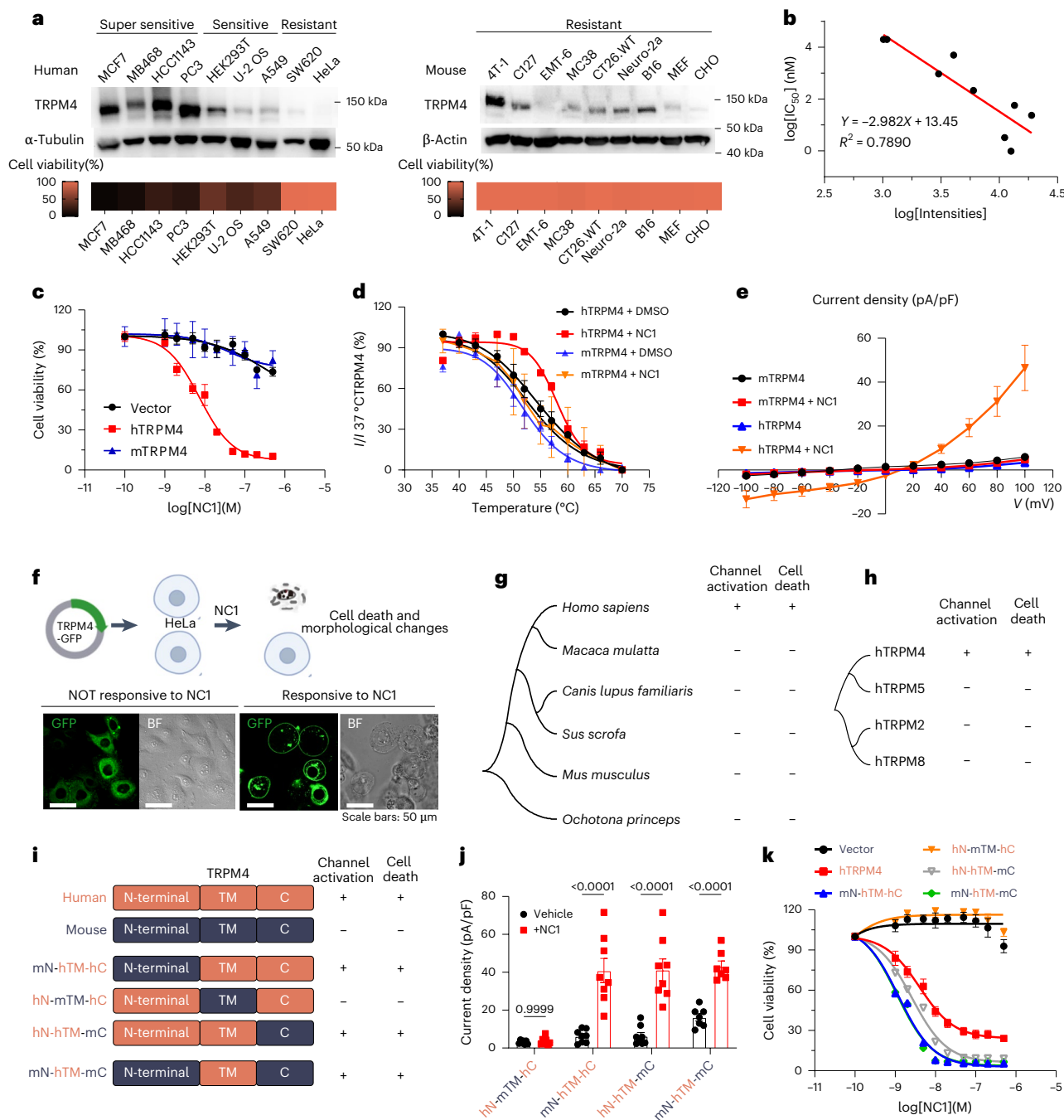


Fig. 3 | NC1 specifically targets hTRPM4 through interaction with the TM region. **a**, TRPM4 protein expression profiles for human (left) and mouse (right) cell lines and corresponding heat map for sensitivity to NC1-induced cell death. **b**, Correlation of TRPM4 expression and NC1 sensitivity of human cell lines. Each dot represents one human cell line. TRPM4 band intensities of the indicated samples from **a** were analyzed using ImageJ software. IC_{50} values of NC1 were from Extended Data Fig. 1c. The linear regression was fitted with GraphPad Prism 9.0.0. **c**, Dose-dependent cytotoxicity assay of NC1 in TRPM4-KO MCF7 cells with re-expression of hTRPM4 or mTRPM4. **d**, CETSA of NC1 in HEK293T hTRPM4 or mTRPM4 cells. The graph represents the relative TRPM4 band intensities of the indicated samples as the mean $\pm$ s.d. from three independent experiments. **e**, The whole-cell current density–voltage relation of hTRPM4 or mTRPM4 expressed in HeLa cells. **f**, Schematic diagram of the validation system to verify whether mutant or chimera TRPM4 responds to NC1. **g**, Left: phylogenetic tree constructed by CLUSTALW multiple-sequence alignments for TRPM4 proteins (with accession numbers of protein sequences) from several species, namely, *Homo sapiens* (NP_060106.2), *Macaca mulatta* (XP_028695988.1), *Canis lupus familiaris* (XP_541500.3),

Sus scrofa (XP_020950369.1), *Mus musculus* (NP_780339.2) and *Ochotona princeps* (XP_012785770.1) isoforms of TRPM4. Right: summary results of their corresponding sensitivity to NC1-induced channel activation or cell death. **h**, Phylogenetic tree of TRPM family protein and summary results of their corresponding sensitivity to NC1. **i–k**, TRPM4 chimeric constructs and their corresponding sensitivity to NC1. hTRPM4 and mTRPM4 are indicated by h and m, respectively. Chimeric constructs were generated by PCR cloning (**i**). Statistics showing the sensitivity to NC1-induced current densities of TRPM4 chimeras at +100 mV in whole-cell patch-clamp experiments using HeLa cells (**j**). NC1-induced cell death in cell cytotoxicity experiments using MCF7 TRPM4-KO cells respectively re-expressed with hTRPM4 or chimeric TRPM4 constructs as shown (**k**). Viability data represent the mean $\pm$ s.d. of $n = 3$ replicates from one representative of three independent experiments. Electrophysiological data represent the mean $\pm$ s.e.m. of ($n = 6$ for hTRPM4 in **e**; $n = 4$ for mTRPM4 in **e**; $n = 7$ in **j**) independent cells. Micrograph and immunoblot data show one representative of three replicates from three independent experiments. Statistical analysis was performed using a two-way ANOVA (GraphPad Prism 9.0.0) in **j**.

for NC1 sensitivity. Both hTRPM4 and mTRPM4 were divided into three fragments: N terminus (1–767 in human, 1–763 in mouse), transmembrane (TM) (768–1093 in human, 764–1089 in mouse) and C terminus (1094–1214 in human, 1090–1213 in mouse) (Fig. 3i and Extended Data Fig. 5a). The TM fragment of TRPM4 includes the pre-S1 (S0) domain, S1–S6 core TM domains and the TRP domain embedded in the lipid layer (Extended Data Fig. 5a). Swapping the hTRPM4 TM region with the mTRPM4 TM region blocked NC1 sensitivity (hN-mTM-hC), while replacing the N (mN-hTM-hC) or C (hN-hTM-mC) terminus retained sensitivity (Fig. 3i–k, Extended Data Fig. 5b,c and Supplementary Fig. 4a–d). Replacing the mTRPM4 TM region with the hTRPM4 TM (mN-hTM-mC) restored NC1 sensitivity (Fig. 3i–k and Supplementary Fig. 4a–d). NC1-sensitive chimeras fully rescued NECSO phenotypes in TRPM4-KO MCF7 cells (Fig. 3k), confirming the hTRPM4 TM region's critical role in NC1 sensitivity.

Molecular docking identifies key residues for NC1 activation

Domain-swapping analysis highlighted the importance of the TM region of hTRPM4 in NC1 sensitivity. Molecular docking using the hTRPM4 structure (Protein Data Bank (PDB) 6BQV) revealed that NC1 binds within a cavity in the TM region. This cavity is surrounded by the S3 and S4 helices, the S4–S5 linker from one subunit and the S5 and S6 helices from an adjacent subunit. Known as the vanilloid-binding pocket (VBP), this region is a common ligand-gating site in TRP channels²⁷ but is here reported as a TRPM4 ligand-binding site (Fig. 4a and Extended Data Figs. 6 and 7). Docking predicts that NC1 interacts with key residues in this interfacial cavity near the Ca²⁺-binding site, including F936 (S5), W864 (S3) and V904, L907 and H908 (S4) (Fig. 4b,c), suggesting its role in NC1 gating. H908 forms hydrogen bonds and π – π stacking with NC1's indole and hydroxyphenyl groups, while W864 forms hydrogen bonds with the hydroxyphenyl moiety. V904, L907 and F936 stabilize NC1's methylindoline group through hydrophobic interactions (Fig. 4b,c, Extended Data Figs. 6–8, Supplementary Video 2 and Supplementary Data 1). To confirm the importance of this pocket, we substituted key residues. Substitution of W864, H908 or F936 with alanine disrupted NC1 binding, TRPM4 channel activation and NECSO sensitivity (Fig. 4d–g). Substitution of M927 to arginine blocked NC1 access and abolished its effects, while M927A retained NC1 responsiveness (Fig. 4e,f). Electrophysiological analysis showed that these mutants (W864A, H908A and F936A), unresponsive to NC1, retained varying ion permeability compared to the nonfunctional mutant D984A (Fig. 4h). These findings confirm the predicted pocket's essential role in NC1-induced TRPM4 activation.

Molecular dynamics (MD) simulations modeled the NC1–TRPM4 complex in a phospholipid bilayer (Extended Data Fig. 8a). The root-mean-square deviation (r.m.s.d.) stabilized under 5 Å within 100 ns, confirming a stable complex. Interaction energy analysis validated NC1's stable binding to TRPM4 (Extended Data Fig. 8b–d). Binding energy contributions revealed key stabilizing residues: S863, W864, H908 and F936, consistent with molecular docking predictions (Extended Data Fig. 8e). These findings suggest that a VBP-like cavity participates in NC1 activation of TRPM4.

The TM region of TRPM4 includes six TM helices and a TRP helix. The S5–S6 helices and P-loop form the ion-conducting pore, surrounded by the S1–S4 domain, which contains a Ca²⁺-binding site. The S4–S5 linker connects these regions and is crucial for channel gating. While TRPM4 is impermeable to Ca²⁺, it is activated by intracellular Ca²⁺ (refs. 20,28). To determine whether NC1 affects channel activity through Ca²⁺, we tested its effects with varying EGTA (Ca²⁺ chelator) concentrations. NC1 activated TRPM4 independently of Ca²⁺ (Extended Data Fig. 9a–d). Compared to calcium, NC1 strongly augmented TRPM4 currents and intracellular Ca²⁺ did not further enhance NC1-induced activation (Extended Data Fig. 9e). Additionally, calcium chelation did not inhibit NC1-induced NECSO (Extended Data Fig. 9f,g).

Point substitutions at the calcium-binding site^{23,29}, E828K and E1068Q, were tested for NC1 responsiveness. E828K completely blocked NC1-induced activation, while E1068Q retained reduced

responsiveness (Extended Data Fig. 9h,i). In TRPM4-KO cells, these mutants showed partial or no NECSO-inducing effects (Extended Data Fig. 9j). These findings suggest that NC1 and calcium may share common mechanisms in activating TRPM4 channel activity.

Energy-depletion-induced necrosis shares features of NECSO

Dysregulation of TRPM4 is linked to ATP depletion in conditions such as ischemia and spinal cord injury^{30,31}. We hypothesized that NC1-induced NECSO mimics energy-depletion-induced necrosis. Cos7 cells expressing WT TRPM4 or mutants (H159A with reduced ATP inhibition²⁴, channel-deficient D984A and E1068Q with reduced calcium sensitivity) were tested for cell death through LDH release assay upon NC1 exposure or 2-deoxy-D-glucose (2DG) + NaN₃ (energy depletion). WT TRPM4 cells showed dose-dependent cell death with NC1 and energy depletion, while H159A cells were more vulnerable to NC1 (Fig. 5a). Conversely, D984A and E1068Q mutants exhibited minimal cell death under either condition. NC1-binding defective mutants (W864A, H908A and F936A) also resisted energy-depletion-induced cell death (Fig. 5b). These findings suggest that energy-depletion-induced necrosis shares a genetic profile with NECSO.

We examined NECSO under energy depletion. In Cos7 cells with hTRPM4, 2DG activated TRPM4 channel activity (Fig. 5c). Energy depletion triggered sodium influx and necrotic cell death, both counteracted by sodium depletion with NMDG (Fig. 5d,e). These results indicate that energy depletion may also induce NECSO.

GOF mutants of hTRPM4 show NECSO sensitivity

TRPM4 mutations are linked to cardiac arrhythmias, including progressive familial heart block type 1, Brugada syndrome and long QT syndrome, which can lead to sudden cardiac death (SCD)^{32–34}. Over 25 mutations have been associated with familial cardiac diseases, with some acting as gain-of-function (GOF) and others as loss-of-function (LOF) mutants. Abnormal energy supply or stress could trigger SCD in these populations (Supplementary Table 1).

TRPM4 is more abundant in Purkinje fibers, which are involved in ventricular arrhythmias³⁵. We hypothesized that disease-related mutations may alter NECSO sensitivity in heart cells, leading to injury. Several GOF and LOF TRPM4 mutants were expressed in rat H9C2 cardiomyoblasts. GOF mutants showed increased sensitivity to NECSO, while LOF mutants did not respond (Fig. 5f and Supplementary Table 1). This suggests that TRPM4 GOF mutations may enhance cell death and injury during energy-depletion conditions. Consistently, TRPM4 KO protected human cardiomyoblasts from NC1 treatment and energy depletion (Supplementary Fig. 5a,b). Additionally, overexpression of TRPM4 but not other TRPM family members was upregulated in left-ventricular samples from people with heart failure across 16 transcriptomic studies (Fig. 5g).

Identification of CLT and DHPs as NECSO inhibitors

To identify small-molecule inhibitors of NECSO, we used in silico screening with the newly identified NC1-binding pocket of TRPM4. Clotrimazole (CLT), an antifungal drug, was found to share a similar structure to NC1 and fit well into the binding cavity. Molecular docking showed that CLT binds through hydrophobic interactions, van der Waals contacts and hydrogen bonds, interacting with residues such as M927 in S5, F931 in S6 and H908 in S4 (Fig. 6a and Supplementary Data 2). These interactions suggest that CLT competes with NC1 to inhibit TRPM4 and NECSO. Testing in MCF7 cells confirmed CLT's potency, with an EC₅₀ of 4 μ M (Fig. 6b). Its inhibitory effect on TRPM4 was also confirmed through electrophysiology (Fig. 6c and Supplementary Fig. 6).

In the second strategy, we screened a US Food and Drug Administration-approved drug library of 1,376 molecules to identify inhibitors of NC1-induced cell death in MCF7 cells. The nonspecific TRPM4 antagonist glibenclamide³⁶ and mitochondrial permeability transition pore inhibitor cyclosporine A (CsA)³⁷ were selected, validating

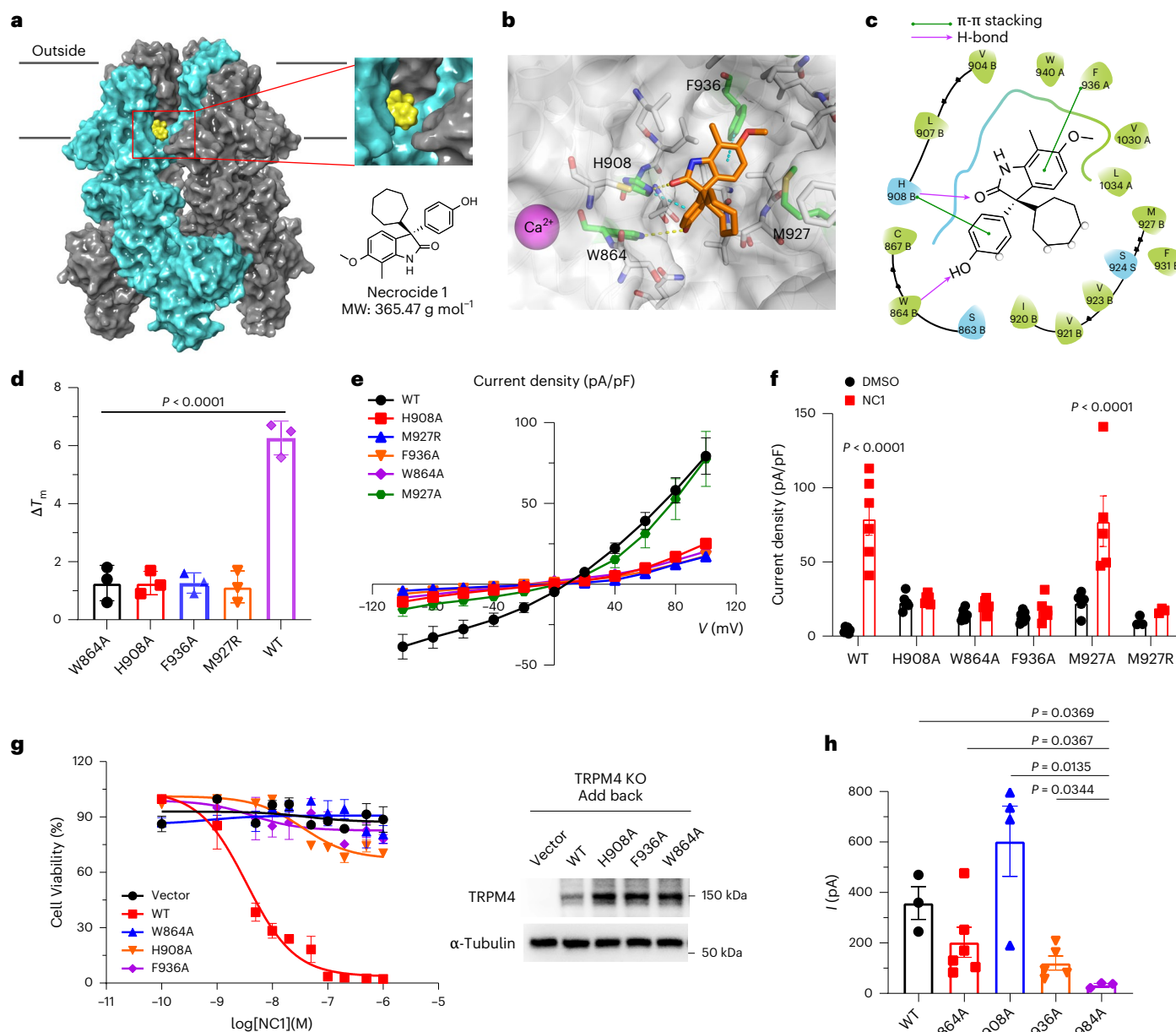


Fig. 4 | Docking and mutagenesis analysis reveal critical residues of TRPM4 for NC1 activation. **a**, Surface representation showing the overall structure of the molecular modeling TRPM4–NC1 complex. MW, molecular weight. Inset: a close-up view of the NC1-binding site. Individual TRPM4 subunits are colored cyan or gray, whereas NC1 is highlighted with yellow. Lower right: chemical structure of NC1. **b**, **c**, Specific recognition of NC1-bound hTRPM4. A 3D diagram showing that NC1 binds a deep cavity of TRPM4 between S3, S4, S5' and S6' helix; Ca^{2+} is colored purple (**b**). A 2D diagram showing that NC1 interacts with the key residues of TRPM4 through a mixture of hydrogen bonds, hydrophobic interactions and π - π stacking (**c**). **d**, **e**, CETSA of NC1 and TRPM4 mutants of key residues in **b**, **c**. $\Delta T_m = T_{m(\text{NC1})} - T_{m(\text{DMSO})}$. Melting temperatures (T_m) were calculated by TRPM4 band intensities of the indicated samples in western blot with GraphPad. The graph represents mean $\pm$ s.d. from three independent experiments. **e**, The current density–voltage relationship of hTRPM4 mutants involving key residues for NC1 binding was examined in HeLa cells. **f**, Summary plot showing the maximum

amplitude of current densities activated by NC1 in the TRPM4 WT channel compared to the aforementioned mutants. **g**, Dose-dependent cytotoxicity assay of NC1 in TRPM4-KO MCF7 cells with re-expression of WT TRPM4 or TRPM4 mutants (W864A, H908A, F936A). **h**, Summary data showing the constitutive activity of hTRPM4 mutants (W864A, H908A, F936A) recorded from inside-out patches at -100 mV. WT TRPM4 was shown as positive control and D984A as negative control. Viability data represent the mean $\pm$ s.d. of $n = 3$ replicates from one representative of three independent experiments. Electrophysiological data represent the mean $\pm$ s.e.m. of ($n = 6$ for WT, H908A and W864A, 7 for F936A, 4 for M927A and 3 for M927R in **e**, **f**; $n = 3$ for WT, D984A and M927R, 4 for H908A, 5 for F936A and 6 for W864A in **h**) independent cells. Immunoblot data show one representative of three replicates from three independent experiments. Statistical analysis was performed using a one-way ANOVA (GraphPad Prism 9.0.0) for **d**, **h** and two-way ANOVA (GraphPad Prism 9.0.0) for **f**.

the screening strategy (Fig. 6d). Dihydropyridines (DHPs), known calcium-channel blockers targeting L-type voltage-gated calcium channels (LTCCs)³⁸, also showed strong NECO inhibitory effects in MCF7 cells (Fig. 6b). We hypothesized that DHPs and CLT block NECO through different mechanisms; DHPs indirectly suppress NC1-mediated

TRPM4 activation, while CLT directly inhibits TRPM4. Electrophysiological results confirmed that DHPs did not affect TRPM4 channel activity but CLT directly inhibited it (Fig. 6c). Both DHPs and CLT blocked sodium influx induced by NC1 (Fig. 6e). Moreover, we tested whether other calcium channels were involved in NECO and found that only LTCC

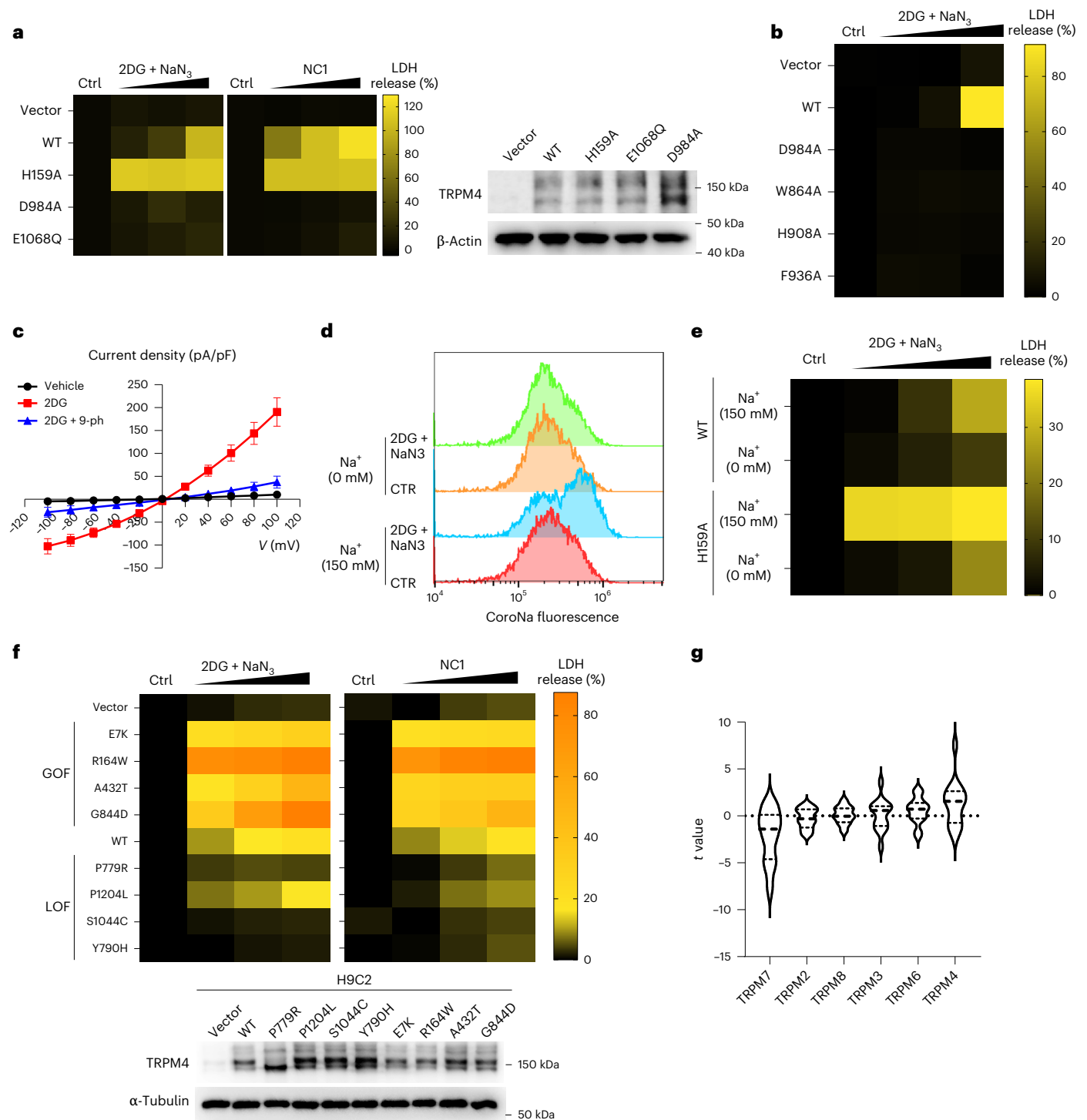


Fig. 5 | Energy-depletion-induced cell death shares features of NECSO. a, Left: heat map depicting the cell death (measured by LDH release) of Cos7 cells stably expressing various hTRPM4 mutants and then treated with 2DG + NaN₃ (0, 5, 10 or 20 mM 2DG plus 1 mM NaN₃) or NC1 (0, 200, 500 or 1,000 nM) for 16 h. **Right:** immunoblot analysis of the expression levels of different hTRPM4 mutants in Cos7 cells. **b,** Heat map depicting the cell death of Cos7 cells expressing various NCI-insensitive hTRPM4 mutants and then treated with 2DG + NaN₃. **c,** The whole-cell current density–voltage relation of Cos7 cells expressing hTRPM4 activated by 2DG. **d,** Intracellular sodium levels measured with CoroNa in Cos7 cells stably expressing hTRPM4 treated with vehicle or 20 mM 2DG plus 1 mM NaN₃ in EBS for 6 h by fluorescence-activated cell sorting (FACS). The sodium in EBS was either maintained at physiological concentration (150 mM) or completely replaced with equal molar NMDG. **e,** Heat map illustrating the cell death of Cos7 cells transfected with WT hTRPM4 or H159A mutant and

subsequently treated with 2DG + NaN₃ under sodium-free and sodium-containing conditions. **f,** Top: heat map depicting the cell death of rat cardiac myoblast H9C2 cells transiently expressing various GOF or LOF mutants of hTRPM4 and then treated with 2DG + NaN₃ or NC1. Bottom: immunoblot analysis of the expression levels of different hTRPM4 mutants in H9C2 cells. **g,** The expression levels of TRPM4 in human failing heart according to a meta-analysis of multiple studies (<https://saezlab.shinyapps.io/reheat/>). The *t* value represents the corrected gene expression level, with each data point representing an independent study. The bold dashed line represents the mean value, while the thin dashed lines represent the upper and lower quartiles. Cell death data represent the mean of *n* = 3 replicates from one representative of three independent experiments. Electrophysiological data represent the mean ± s.e.m. of *n* = 6 independent cells. Flow cytometry and immunoblot data show one representative of three replicates from three independent experiments.

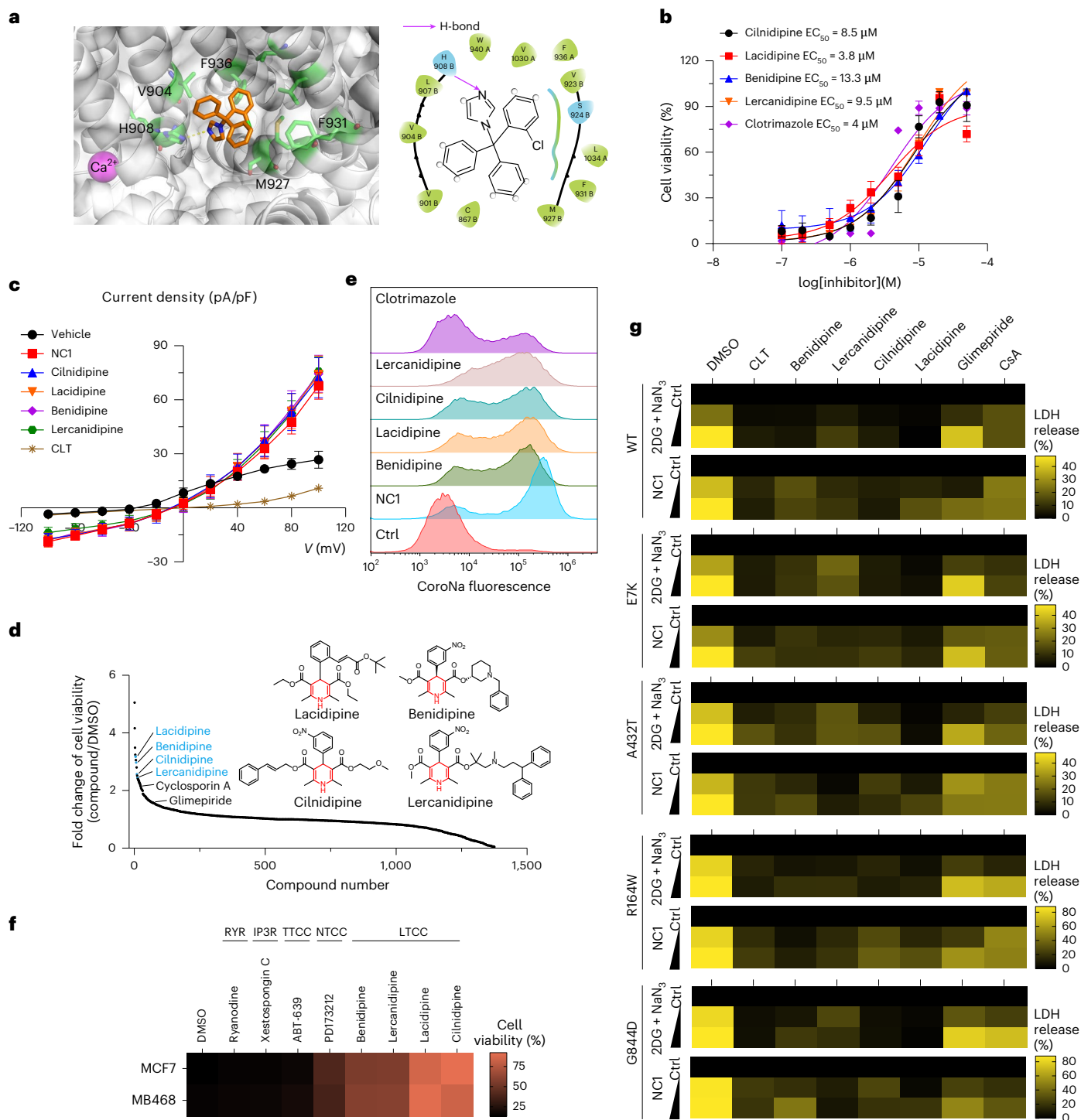


Fig. 6 | DHPs and CLT function as small-molecule inhibitors of NECO. a, Left: 3D diagram depicting that CLT specifically binds to hTRPM4, recognizing a deep cavity similar to NC1. **Right:** 2D diagram showing that CLT interacts with the key residues of TRPM4 through a mixture of hydrogen bonds and hydrophobic interactions. **b,** CLT and DHPs presented an inhibitory effect on NC1-induced cell death. MCF7 cells were pretreated with increasing concentrations of CLT or DHPs followed by 100 nM NC1 treatment and cell viability was assessed using CTG 24 h thereafter. **c,** The whole-cell current density–voltage relation of HEK293T cells stably expressing hTRPM4. Cells were pretreated with 30 μ M DHPs or CLT followed by 1 μ M NC1 treatment. **d,** Top hits of inhibitor screening for NECO. The DHP dots are highlighted in cyan color. Insets: chemical structures. **e,** Intracellular sodium levels measured by FACS using CoroNa in

MCF7 cells pretreated with 30 μ M DHPs or CLT for 1 h and then exposure to 100 nM NC1 for 3 h. **f,** Heat map illustrating the cell viability of MCF7 or MB468 cells pretreated with various types of calcium-channel inhibitors followed by NC1 exposure. **g,** Heat map depicting the cell death measured by LDH release of H9C2 cells expressing hTRPM4 mutants treated with 2DG + NaN_3 or NC1 combined with treatment of different inhibitors. Cell death data represent the mean of $n = 3$ replicates from one representative of three independent experiments. Electrophysiological data represent the mean $\pm$ s.e.m. of ($n = 4$ for vehicle, NC1 and CLT and 3 for DHPs) independent cells. Flow cytometry and cell death data show one representative of three replicates from three independent experiments.

blockers efficiently inhibited NECSO in MCF7 and MB468 cells (Fig. 6f). LTCCs are activated by TRPM4-dependent membrane depolarization³⁹, suggesting a feedback loop between these channels regulating NECSO. The above data demonstrate that DHPs inhibit NECSO by targeting LTCCs to prevent TRPM4 activation, while CLT blocks NECSO by directly competing with NC1 for its binding pocket on TRPM4.

DHPs and CLT were tested for their ability to block energy-depletion-induced cell death. Both inhibitors reduced NC1 and energy-depletion-induced NECSO more effectively than glimepiride in Cos7 cells (Extended Data Fig. 10). They also protected cardiomyoblasts expressing TRPM4 GoF mutants from energy depletion (Fig. 6g). These findings suggest that NECSO inhibitors could have therapeutic potential to protect cardiac cells, particularly in conditions with TRPM4 GOF mutations.

Discussion

In this study, we reported that a small-molecule compound elicits necrotic cell death, which is characterized by sodium overload and dependent on monovalent cation channel TRPM4. The small-molecule compound NC1 serves as a highly selective agonist for TRPM4 through a pocket architecturally akin to the ligand-binding regulatory pocket found in other TRP family proteins. NC1 persistently activates TRPM4 channel activity. Consequently, uncontrolled opening of TRPM4 channel causes massive sodium influx and membrane depolarization, leading to necrotic cell death, which we termed as NECSO to separate from other types of necrotic cell death. NECSO could be blocked by calcium-channel blocker indirectly and an antifungal drug CLT as a direct competitor. Energy depletion likely activates NECSO given its shared cell morphology change, genetic and chemical intervention profiles. Cardiac arrhythmias associated with TRPM4 GOF mutants are more vulnerable to NECSO and can be blocked by NECSO inhibitors.

TRPM4 has been implicated in cellular demise in diverse cellular contexts, including neurons^{31,40,41}, astrocytes^{19,42} and endothelial cells^{30,42}. This demise has long been classified as oncosis⁴³. However, the oncosis classification presents an obscure definition based on morphological changes such as cell swelling and membrane blebbing that often result from cellular stress or damage leading to cell membrane permeability changes. There are many factors that might cause cell membrane permeability changes and, in this study, we described NECSO as a type of necrotic cell death wherein necrotic cell death is incited specifically by the surge of sodium ions mediated by the activation of TRPM4. Although persistent TRPM4 activation might also lead to the influx of monovalent cations such as lithium and cesium, our study highlights sodium influx as the pivotal driver of cellular demise resulting in the most pronounced cell death response. Moreover, the influx of sodium rather than other cations mediated by TRPM4 likely governs the major cellular responsiveness to NC1 *in vivo*, given the limited physiological concentrations of other monovalent cations in extracellular fluid.

NECSO stands apart in its genetic and biochemical composition, diverging markedly from apoptosis, autophagy and various well-defined programmed necrosis pathways. NECSO likely features a more restrained level of membrane permeability, selectively permitting ion exchange before extensive membrane rupture. The precise mechanism by which sodium influx triggers necrosis remains unclear, although membrane depolarization is thought to bear importance. The participation of the sodium pump ($\text{Na}^+ - \text{K}^+$ ATPase) in this process is contemplated. Excessive sodium influx would drive hyperactivation of sodium pumps, which relentlessly hydrolyze ATP to expel sodium from the cell, thus maintaining membrane potential homeostasis. This process, intensively energy consuming, could demand up to 70% of cellular energy, particularly notable in certain neural tissues⁴⁴. Continuous hyperactivation of sodium pumps would deplete the cellular ATP reservoir, leading to energy exhaustion. Supporting this notion, our observations underscore the influence of mitochondria on NECSO, where CsA, a mitochondrial permeability transition inhibitor, effectively curtails NC1-triggered cell

demise⁴⁵. Notably, mitochondria's roles in ATP or reactive oxygen species generation could potentially intersect with NECSO regulation, given their documented modulation of TRPM4 channel activity^{19,45,46}. Additionally, excessive sodium might directly impede mitochondrial energy production, as Na^+ operates as a secondary messenger modulating the function of the mitochondrial oxidative phosphorylation system by influencing inner mitochondrial membrane fluidity⁴⁷. These hypotheses will be subjected to further exploration in upcoming investigations.

TRPM4 is a member of the TRP channel family, complex poly-modal signal sensors adept at integrating information from physical and/or chemical stimuli to modulate cellular excitability across diverse environmental or physiological contexts⁴⁸. Different from extensively explored TRPV and TRPA families, which serve as receptors for vital sensory perceptions such as heat, pain and cold, the function and regulatory mechanism of TRPM4 remain largely enigmatic because no natural or physiological ligand of TRPM4 has been reported until now. Compounds such as intracellular calcium, decavanadate, tissue plasminogen activator, U73122 and diazoxide have been documented to activate TRPM4 (ref. 49), likely different from NC1. NC1 potently and persistently activates TRPM4 channel activity and retains its membrane localization to enable a robust influx of sodium ions. Molecular docking has revealed that a cavity in the TM region surrounded by the S3 and S4 helices and S4–S5 linker from one subunit and S5 and S6 helix from the adjacent subunit is crucial for NC1 effect. This pocket exhibits a degree of conservation across TRP channels, which is often referred as the VBP pocket for ligand binding. For example, ligand–receptor pairs including vanilloid–TRPV1 (PDB 5IRX), ECN–TRPV5 (PDB 6B5V) and GNE551–TRPA1 (PDB 6X2J) all bind through VBPs. NC1 is the first chemical ligand shown to activate TRPM4 through the VBP region, which might explain its potent activation of this channel. The docking results may have limitations because the molecular docking was conducted on the basis of the published calcium-bound structure of hTRPM4, which is still a closed state²³. However, upon NC1 binding, TRPM4 most likely transformed to an open state and was trapped in that conformation by NC1. The compound might eventually penetrate into a more embedded reservoir and, thus, can no longer be washed out, which can be confirmed through further structural analysis. We also noticed a significant difference between the EC_{50} of NC1 in cells and the K_d value obtained from the *in vitro* SPR experiment. We detected a K_d value of 880 nM for the NC1–TRPM4 interaction, with no significant binding observed for NC1i (Fig. 1f and Supplementary Fig. 2e,f). However, the K_d value still exhibits some disparity with the EC_{50} of NC1. We suspect the lipid binding and likely protein binding of TRPM4 might contribute to this difference. TRPM4 is a TM protein and lipids (phosphatidylinositol 4,5-bisphosphate) on the cell membrane can bind to TRPM4, thus affecting its conformational stability or activation^{24,50}. In SPR experiments, TRPM4 is directly coupled to the chip without lipid binding. Thus, the K_d value might not be as low as the EC_{50} in the cellular circumstance. Some TRPM4-interacting proteins might also help the binding of TRPM4 with NC1. All these possibilities will be systemically and thoroughly studied in the future.

Interestingly, NC1 specifically activates hTRPM4 but not mTRPM4 despite the fact that hTRPM4 and mTRPM4 share 88% sequence homology. NC1 failed to bind mTRPM4 and mTRPM4 is unable to rescue the deficiency caused by hTRPM4 KO. We demonstrated that the TM domain accounts for the distinct sensitivity to NC1 in hTRPM4 and mTRPM4. In addition, NC1 might stabilize and activate TRPM4 allosterically, which might help to capture dynamic channel activation processes. Structural studies on TRPV1 with double-knot toxin, a peptide spider toxin stabilizing the open state of TRPV1, successfully revealed the dynamic regulation of TRPV1 (ref. 51). Until now, the structure of the open state of TRPM4 has yet to be elucidated^{23,24} and future studies on the structure of NC1–hTRPM4 might help to resolve this problem. Furthermore, the NC1-binding pocket might be a valuable target for drug design for TRPM4 agonists and antagonists, which is illustrated

by the example of CLT, an antifungal agent sharing structural similarities with NC1 and fitting in the NC1–hTRPM4 pocket. CLT serves as an antagonist to restrain NC1-triggered NECSO. CLT is the ring-opened analog of NC1 with a similar three-dimensional (3D) structure containing a combination of hydrophobicity and π – π interactions, a spherical shape and an all-carbon quaternary stereocenter. CLT's inhibitory effect likely occurs through competition with NC1 because both compounds share a common pocket within TRPM4. The inhibitory effect of CLT on TRPM4 and other TRPM proteins has also been reported by other groups^{49,52}. Notably, CLT's neuroprotective impact on spinal cord ischemia–reperfusion injury⁵³ likely also arises from TRPM4 inhibition.

Under physiological circumstances, the propensity for TRPM4 to open is heightened by elevated cytoplasmic $[Ca^{2+}]$ and diminished ATP concentrations. Augmented cytosolic Ca^{2+} levels or profound ATP depletion often triggers TRPM4-dependent cell death in pathological contexts⁵⁴. Consistently, we observed that energy-depletion-induced necrotic cell death shares largely overlapped morphology and chemical profiles with NECSO, suggesting that NECSO is involved in energy-depletion-induced necrotic cell death. Furthermore, TRPM4 dysfunctions have been implicated in a broad spectrum of pathological contexts involving cell and tissue injuries including stroke, brain or spinal cord injury, infection, cardiac diseases and cancers^{31,40,41,55}.

In this study, we observed that cardiac disease-associated TRPM4 GOF mutations are more vulnerable to NECSO and the inhibitors are able to block the resulting cell death. It is reported that TRPM4 not only has an essential role in cardiac conduction but is also involved in hypertension, hypertrophy and heart failure^{56,57}. TRPM4 KO improves survival rate and enhances β -adrenergic cardiac reserve after ischemic heart failure and TRPM4 inhibitor 9-phenanthrol was also found to reduce ischemia–reperfusion-induced cardiac death in rats⁵⁸. An increase in expression of TRPM4 in human failing heart tissue was also reported⁵⁹. Previous investigations of cardiac diseases mostly focused on the electrophysiological function of TRPM4. Our research supplies a mechanism underlying the function of TRPM4 as a cell death mediator. Therefore, development of TRPM4 antagonists might hold promise to protect cells from sodium-overload-associated cell and tissue injuries. On the other hand, NECSO inducers such as NC1 might represent a promising avenue for cancer therapy. TRPM4 dysfunction has been linked to multiple cancers and NC1 derivatives have been shown to present robust anticancer effects on xenograft tumor models⁶⁰. However, how to balance the killing effects of TRPM4 in normal and cancer cells remains a challenge. Nevertheless, these studies implicate the involvement of NECSO in a broad spectrum of human diseases, the targeting of which might bring beneficial effects to populations if we carefully elucidate the detailed mechanisms and tease apart the critical positive and unwanted side effects in each pathological condition. With the critical genes and chemical tools identified in this study, we can start to address the biological importance of NECSO.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41589-025-01841-3>.

References

- Thornton, S. N. Sodium intake, cardiovascular disease, and physiology. *Nat. Rev. Cardiol.* **15**, 497 (2018).
- Friedman, P. A. Codependence of renal calcium and sodium transport. *Annu. Rev. Physiol.* **60**, 179–197 (1998).
- Amiry-Moghaddam, M. & Ottersen, O. P. The molecular basis of water transport in the brain. *Nat. Rev. Neurosci.* **4**, 991–1001 (2003).
- Hunter, R. W., Dhaun, N. & Bailey, M. A. The impact of excessive salt intake on human health. *Nat. Rev. Nephrol.* **18**, 321–335 (2022).
- Liang, D., Bhatta, S., Gerzanich, V. & Simard, J. M. Cytotoxic edema: mechanisms of pathological cell swelling. *Neurosurg. Focus* **22**, E2 (2007).
- Rungta, R. L. et al. The cellular mechanisms of neuronal swelling underlying cytotoxic edema. *Cell* **161**, 610–621 (2015).
- Wang, Y. et al. Brain tissue sodium is a ticking clock telling time after arterial occlusion in rat focal cerebral ischemia. *Stroke* **31**, 1386–1391 (2000).
- Papadopoulos, M. C. et al. Molecular mechanisms of brain tumor edema. *Neuroscience* **129**, 1011–1020 (2004).
- Dixon, S. J. et al. Ferroptosis: an iron-dependent form of nonapoptotic cell death. *Cell* **149**, 1060–1072 (2012).
- Tsvetkov, P. et al. Copper induces cell death by targeting lipoylated TCA cycle proteins. *Science* **375**, 1254–1261 (2022).
- Cai, Z. et al. Plasma membrane translocation of trimerized MLKL protein is required for TNF-induced necroptosis. *Nat. Cell Biol.* **16**, 55–65 (2014).
- Boron, W. F. & Boulpaep, E. L. *Medical Physiology: A Cellular and Molecular Approach* 2nd edn (Saunders, 2012).
- Kadir, L. A., Stacey, M. & Barrett-Jolley, R. Emerging roles of the membrane potential: action beyond the action potential. *Front. Physiol.* **9**, 1661 (2018).
- Murphy, E. & Eisner, D. A. Regulation of intracellular and mitochondrial sodium in health and disease. *Circ. Res.* **104**, 292–303 (2009).
- Zhang, J. et al. Necrocytotoxic 1 mediates necrotic cell death and immunogenic response in human cancer cells. *Cell Death Dis.* **14**, 238 (2023).
- Wang, B. et al. Integrative analysis of pooled CRISPR genetic screens using MAGeCKFlute. *Nat. Protoc.* **14**, 756–780 (2019).
- Grand, T. et al. 9-Phenanthrol inhibits human TRPM4 but not TRPM5 cationic channels. *Br. J. Pharmacol.* **153**, 1697–1705 (2008).
- Nilius, B. et al. The selectivity filter of the cation channel TRPM4. *J. Biol. Chem.* **280**, 22899–22906 (2005).
- Chen, M. & Simard, J. M. Cell swelling and a nonselective cation channel regulated by internal Ca^{2+} and ATP in native reactive astrocytes from adult rat brain. *J. Neurosci.* **21**, 6512–6521 (2001).
- Launay, P. et al. TRPM4 is a Ca^{2+} -activated nonselective cation channel mediating cell membrane depolarization. *Cell* **109**, 397–407 (2002).
- Bohlen, C. J. et al. A bivalent tarantula toxin activates the capsaicin receptor, TRPV1, by targeting the outer pore domain. *Cell* **141**, 834–845 (2010).
- Lin King, J. V. et al. A cell-penetrating scorpion toxin enables mode-specific modulation of TRPA1 and pain. *Cell* **178**, 1362–1374 (2019).
- Autzen, H. E. et al. Structure of the human TRPM4 ion channel in a lipid nanodisc. *Science* **359**, 228–232 (2018).
- Guo, J. et al. Structures of the calcium-activated, non-selective cation channel TRPM4. *Nature* **552**, 205–209 (2017).
- Nilius, B. et al. Regulation of the Ca^{2+} sensitivity of the nonselective cation channel TRPM4. *J. Biol. Chem.* **280**, 6423–6433 (2005).
- Woo, S. K. et al. Complex N-glycosylation stabilizes surface expression of transient receptor potential melastatin 4b protein. *J. Biol. Chem.* **288**, 36409–36417 (2013).
- Cao, E., Liao, M., Cheng, Y. & Julius, D. TRPV1 structures in distinct conformations reveal activation mechanisms. *Nature* **504**, 113–118 (2013).
- Nilius, B. et al. Voltage dependence of the Ca^{2+} -activated cation channel TRPM4. *J. Biol. Chem.* **278**, 30813–30820 (2003).
- Yamaguchi, S., Tanimoto, A., Otsuguro, K., Hibino, H. & Ito, S. Negatively charged amino acids near and in transient receptor potential (TRP) domain of TRPM4 channel are one determinant of its Ca^{2+} sensitivity. *J. Biol. Chem.* **289**, 35265–35282 (2014).

30. Gerzanich, V. et al. De novo expression of Trpm4 initiates secondary hemorrhage in spinal cord injury. *Nat. Med.* **15**, 185–191 (2009).
31. Yan, J., Bengtson, C. P., Buchthal, B., Hagenston, A. M. & Bading, H. Coupling of NMDA receptors and TRPM4 guides discovery of unconventional neuroprotectants. *Science* **370**, eaay3302 (2020).
32. Pironet, A., Vandewiele, F. & Vennekens, R. Exploring the role of TRPM4 in calcium-dependent triggered activity and cardiac arrhythmias. *J. Physiol.* **602**, 1605–1621 (2024).
33. Liu, H. et al. Gain-of-function mutations in *TRPM4* cause autosomal dominant isolated cardiac conduction disease. *Circ. Cardiovasc. Genet.* **3**, 374–385 (2010).
34. Syam, N. et al. Variants of transient receptor potential melastatin member 4 in childhood atrioventricular block. *J. Am. Heart Assoc.* **5**, e001625 (2016).
35. Hof, T. et al. TRPM4 non-selective cation channels influence action potentials in rabbit Purkinje fibres. *J. Physiol.* **594**, 295–306 (2016).
36. Guinamard, R. et al. Functional characterization of a Ca^{2+} -activated non-selective cation channel in human atrial cardiomyocytes. *J. Physiol.* **558**, 75–83 (2004).
37. Halestrap, A. P., Connern, C. P., Griffiths, E. J. & Kerr, P. M. Cyclosporin A binding to mitochondrial cyclophilin inhibits the permeability transition pore and protects hearts from ischaemia/reperfusion injury. *Mol. Cell. Biochem.* **174**, 167–172 (1997).
38. Hess, P., Lansman, J. B. & Tsien, R. W. Different modes of Ca channel gating behaviour favoured by dihydropyridine Ca agonists and antagonists. *Nature* **311**, 538–544 (1984).
39. Gonzales, A. L., Amberg, G. C. & Earley, S. Ca^{2+} release from the sarcoplasmic reticulum is required for sustained TRPM4 activity in cerebral artery smooth muscle cells. *Am. J. Physiol. Cell Physiol.* **299**, C279–C288 (2010).
40. Simard, J. M. et al. Newly expressed SUR1-regulated $\text{NC}_{\text{Ca-ATP}}$ channel mediates cerebral edema after ischemic stroke. *Nat. Med.* **12**, 433–440 (2006).
41. Leiva-Salcedo, E., Riquelme, D., Cerda, O. & Stutzin, A. TRPM4 activation by chemically- and oxygen deprivation-induced ischemia and reperfusion triggers neuronal death. *Channels (Austin)* **11**, 624–635 (2017).
42. Wei, S. et al. Comparison of anti-oncotic effect of TRPM4 blocking antibody in neuron, astrocyte and vascular endothelial cell under hypoxia. *Front. Cell Dev. Biol.* **8**, 562584 (2020).
43. Loh, K. Y., Wang, Z. & Liao, P. Oncotic cell death in Stroke. *Rev. Physiol., Biochem. Pharmacol.* **176**, 37–64 (2019).
44. Harris, J. J. & Attwell, D. The energetics of CNS white matter. *J. Neurosci.* **32**, 356–371 (2012).
45. Nilius, B., Prenen, J., Voets, T. & Droogmans, G. Intracellular nucleotides and polyamines inhibit the Ca^{2+} -activated cation channel TRPM4b. *Pflugers Arch.* **448**, 70–75 (2004).
46. Simon, F. et al. Hydrogen peroxide removes TRPM4 current desensitization conferring increased vulnerability to necrotic cell death. *J. Biol. Chem.* **285**, 37150–37158 (2010).
47. Hernansanz-Agustin, P. et al. Na^+ controls hypoxic signalling by the mitochondrial respiratory chain. *Nature* **586**, 287–288 (2020).
48. Diver, M. M., Lin King, J. V., Julius, D. & Cheng, Y. Sensory TRP channels in three dimensions. *Annu. Rev. Biochem.* **91**, 629–649 (2022).
49. Kovacs, Z. M. et al. Pharmacological modulation and (patho) physiological roles of TRPM4 channel—part 1: modulation of TRPM4. *Pharmaceuticals (Basel)* **15**, 81 (2022).
50. Zhang, Z., Okawa, H., Wang, Y. & Liman, E. R. Phosphatidylinositol 4,5-bisphosphate rescues TRPM4 channels from desensitization. *J. Biol. Chem.* **280**, 39185–39192 (2005).
51. Zhang, K., Julius, D. & Cheng, Y. Structural snapshots of TRPV1 reveal mechanism of polymodal functionality. *Cell* **184**, 5138–5150 (2021).
52. Hill, K., McNulty, S. & Randall, A. D. Inhibition of TRPM2 channels by the antifungal agents clotrimazole and econazole. *Naunyn Schmiedeberg's Arch. Pharmacol.* **370**, 227–237 (2004).
53. Usul, H., Arslan, E., Cansever, T., Cobanoglu, U. & Baykal, S. Effects of clotrimazole on experimental spinal cord ischemia/reperfusion injury in rats. *Spine* **33**, 2863–2867 (2008).
54. Simard, J. M., Woo, S. K. & Gerzanich, V. Transient receptor potential melastatin 4 and cell death. *Pflugers Arch.* **464**, 573–582 (2012).
55. Ghosh, S. et al. Plasma membrane channel TRPM4 mediates immunogenic therapy-induced necrosis. *Cancer Res.* **83**, 3115–3130 (2023).
56. Vandewiele, F. et al. TRPM4 inhibition by meclofenamate suppresses Ca-dependent triggered arrhythmias. *Eur. Heart J.* **43**, 4195–419 (2022).
57. Mathar, I. et al. Increased catecholamine secretion contributes to hypertension in TRPM4-deficient mice. *J. Clin. Invest.* **120**, 3267–3279 (2010).
58. Piao, H. L. et al. Transient receptor potential melastatin-4 is involved in hypoxia-reoxygenation injury in the cardiomyocytes. *PLoS ONE* **10**, e0121703 (2015).
59. Flores, R. O. R. et al. Consensus transcriptional landscape of human end-stage heart failure. *J. Am. Heart Assoc.* **10**, e019667 (2021).
60. Boudreau, M. W. et al. A small-molecule activator of the unfolded protein response eradicates human breast tumors in mice. *Sci. Transl. Med.* **13**, eabf1383 (2021).

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Methods

Cell lines and culture conditions

MCF7, MB468, HeLa, H9c2, HEK293T, SW620, C127, CHO, Cos7 and MEF cells were cultured in DMEM containing 4.5 g L⁻¹ glucose and L-glutamine (Sigma-Aldrich). PC3, HCC1143, 4T-1, EMT-6, CT26WT, B16 and MC38 cells were cultured in RPMI-1640 medium containing high glucose, L-glutamine and HEPES (Sigma-Aldrich). U-2OS cells were cultured in McCoy's 5A medium (American Type Culture Collection (ATCC)). A549 cells were cultured in Ham's F-12K medium (BaseMedia). Neuro-2a cells were cultured in Eagle's minimum essential medium (ATCC). AC16 cells were cultured in DMEM:F-12 medium (ATCC). All media were supplemented with 10% FBS (Gibco) and 1% penicillin–streptomycin. The cells were cultured at 37 °C in a humidified atmosphere with 5% CO₂. All cell lines were tested for *Mycoplasma* and were not authenticated.

Plasmids

Plasmid transfections were performed in HeLa or HEK293T cells with PEI (Polysciences) transfection reagent. Virus was produced by cotransfection of HEK293T cells with GAG, POL and pLenti expression plasmids at a 1:1:1 w/w/w ratio, using the PEI transfection reagent. Medium containing secreted virus was obtained after 48 h and sterile-filtered. CRISPR sgRNA sequences targeting TRPM4 were designed using the CRISPR design tool developed by the Zhang laboratory (<http://crispr.mit.edu/>). The oligo sequences preceding the protospacer motif were as follows:

5'-GAACCGCTTCCGCTTGC GCC-3'

5'-CCTAGGGCTCGTTCCCTGCG-3'

5'-GTCAACTATGAACGTCGTGC-3'

Western blotting

Proteins were extracted from the cells using radioimmunoprecipitation assay lysis buffer supplemented with protease and phosphatase inhibitors (APEX BIO). The protein concentration was determined using the BCA protein assay kit (Thermo Fisher Scientific). Equal amounts of protein (20 µg) were separated on 10% SDS–PAGE gels (Bio-Rad Laboratories) and transferred onto PVDF membranes (Bio-Rad Laboratories). The membranes were then blocked in 5% nonfat milk for 1 h at room temperature and incubated overnight at 4 °C with primary antibodies against the target proteins. After washing, the membranes were incubated with horseradish peroxidase-conjugated secondary antibodies for 1 h at room temperature. Protein bands were visualized using an enhanced chemiluminescence substrate and detected using a chemiluminescence imaging system (Bio-Rad Laboratories). The following blotting reagents and antibodies were used anti-α-tubulin (Sigma-Aldrich, T6199; diluted 1:10,000), anti-β-actin (Sigma-Aldrich, A5441; diluted 1:10,000), anti-TRPM4 (Abclonal, A10146; diluted 1:1,000) and anti-Cas9 (Abcam, ab204448; diluted 1:1,000).

Cell viability and death measurement

The cell viability and death were assessed using the CellTiter-Glo (CTG) assay or LDH release assay. Cells were seeded in a 96-well plate at a density of 1 × 10⁴ cells per well and allowed to adhere overnight. For analyses using EBS, culture medium was replaced with isotonic EBS (150 mM NaCl, 5 mM KCl, 10 mM glucose, 10 mM HEPES, 2 mM CaCl₂, 1 mM MgCl₂) just before the experimental treatment. After treatment in various experimental conditions, the culture medium was aspirated. For the CTG assay, the intracellular ATP was measured using the CTG luminescent assay kit (Promega, G7570) according to the manufacturer's instructions as described previously. Briefly, cells were washed with PBS, 100 µl of CTG reagent was added to each well and the plate was gently shaken for 10 min to induce cell lysis and release of intracellular ATP. The luminescence signal, proportional to the amount of ATP present and indicative of cell viability, was measured using a multiwell plate reader. For the LDH release assay, the reaction mixture containing

the culture supernatant and LDH assay solution (Promega) was incubated at room temperature for 30 min, followed by the addition of stop solution. The absorbance was measured at 490 nm using a microplate reader. A standard curve was generated using serially diluted LDH control samples to calculate the percentage of LDH release from treated cells. The experiment was performed in triplicate and the results were expressed as the mean ± s.d.

CRISPR–Cas9-based screening and data processing

A schematic of CRISPR screening is shown in Fig. 2a. Specifically, MCF7 cells stably expressing Cas9 (blasticidin selection) were generated and then infected with a whole-genome CRISPR–Cas9 gRNA lentiviral library (human library, GeCKO version 2.0) that targets 19,050 protein-coding genes with 122,411 gRNA sequences (six targeting gRNA sequences per gene). After being treated with 2 µg ml⁻¹ puromycin for 3 days, the infected cells were referred to as 'day 0'. These cells were subsequently passaged every 2 days for a total of 14 days (denoted as day 14), while maintaining 500-fold coverage. At the end of the 14-day period, the cells were divided into two groups and cultured in medium with DMSO or 2 nM NC1. Samples were collected from the day 0, DMSO and NC1 groups and genomic DNA extraction was performed using a QIAamp DNA blood midi kit (Qiagen). The sgRNA inserts were amplified using NEBNext Q5 hot start HiFi PCR master mix (New England Biolabs). In the second round of PCR, Illumina TruSeq adaptors with i5 and i7 barcodes were added and the PCR products were subjected to next-generation sequencing. The obtained sequencing results were then used for further analysis. The gRNA differences between the DMSO and NC1 groups were calculated and analyzed using the MAGeCKFlute¹⁶.

Electrophysiology

All electrophysiological recordings were obtained using patch-clamp recordings for (1) macroscopic current of TRPM4 and its mutants overexpressed in HEK293T, HeLa or Cos7 cells and (2) membrane potential of MCF7 cells. For transfected cells, patch-clamp recordings were conducted 24–48 h after transfection. For calcium-activated TRPM4 whole-cell recording, the bath solution contained 150 mM NaCl, 10 mM HEPES, 1 mM MgCl₂ and 2 mM CaCl₂, pH 7.3. The pipette solution contained 150 mM KCl, 10 mM HEPES, 5 mM EGTA and 4.45 mM CaCl₂, pH 7.3 (1 µM free Ca²⁺). For the calcium-free condition, 10 mM EGTA was added without CaCl₂. In the inside-out or outside-out patch mode, the extracellular solution contained 150 mM NaCl, 10 mM HEPES, 1 mM MgCl₂ and 2 mM CaCl₂, pH 7.3. The intracellular solution contained 150 mM NaCl, 10 mM HEPES and 5 mM EGTA. The 5 mM EGTA was replaced by 1 mM CaCl₂ for calcium-activated TRPM4 inside-out patch. Patch pipettes were pulled from PC-10 (Narishige) and heat-polished to a resistance of 3–10 MΩ. Patch-clamp recordings were performed with Axopatch-200B amplifier and Axon Digidata 1550A driven by Clampex10 software (Molecular Devices). The current signals were filtered at 1 kHz and digitized at a 10-kHz sampling frequency. The whole-cell currents were recorded by voltage steps from –100 mV to +100 mV in 20-mV increments, followed by stepping down to –100 mV. The series resistance was compensated before forming a tight patch seal. The concentration of free Ca²⁺ was prepared according to MAXCHELATOR. Liquid junction potentials were less than 2 mV, as calculated by JPCalc software. All values were measured using Clampfit 10.6 software.

In the analysis of NC1 activation of TRPM4, the normalized activation current (I/I_0) at +100 mV was fitted with the Hill equation:

$$\frac{I}{I_0} = A_1 + \frac{A_2 - A_1}{1 + 10^{(\log x_0 - x)p}} \quad (1)$$

where I is the current amplitude after application of activators, A_1 and A_2 are constants between 0 and 1, x is the concentration of activator, x_0 is the EC₅₀ and p is the Hill constant.

In the analysis of CLT inhibition of TRPM4, the fraction of unblocked current (I/I_0) at +100 mV was fitted with the Hill equation:

$$\frac{I}{I_0} = A_1 + \frac{A_2 - A_1}{1 + 10^{(\log x_1 - x)p}} \quad (2)$$

where I_0 and I are the current amplitudes before and after the application of inhibitors, A_1 and A_2 are constants between 0 and 1, x is the concentration of the inhibitor, x_1 is the half-maximal inhibitory concentration (IC_{50}) and p is the Hill constant.

ICP-MS

The ICP-MS was conducted according to a previously published paper⁶¹. In brief, after calculation of the cell number, cell pellets were collected directly into the 50-ml metal-free tube and then 2 ml of 69% nitric acid was added. The tubes were placed on the digestion block (DigiPREP Jr., SCP-Science) and the temperature of the block was ramped up to 120 °C and maintained at 120 °C for approximately 1 h until the solution had almost disappeared. Tubes were gently swirled every 20 min during incubation. When the samples had cooled down, 0.5 ml of H_2O_2 was added and the samples were heated at 120 °C for a further 30 min. After digestion was complete, the tubes were allowed to cool to room temperature. The volume was made up to 20 ml with ultrapure water. Aqueous calibration standards were prepared by diluting 1,000 mg L^{-1} ICP-MS standard solutions (Spex CertiPrep) with 10% nitric acid. Elements were detected by ICP-MS (NexION2000G, PerkinElmer). Element concentrations were quantified using external calibration with aqueous standard solutions and ^{103}Rh as an internal standard. A certified liver reference standard (T07243QC, FAPAS) was used for quality control.

Flow cytometry

For measurement of intracellular sodium concentration, cells were detached from culture plates using $1\times$ PBS + EDTA. Single-cell suspensions were prepared in $1\times$ PBS with 4% FBS. The cells were then stained with 2 μM CoroNa-AM dye at 37 °C for 30 min, followed by two washes with $1\times$ PBS 4% FBS. Analysis of stained cells was performed using a CytoFlex S (Beckman Coulter) and postacquisition analysis was conducted using FlowJo software. Sodium concentration were measured under FITC (525/50 505LP). To assess intracellular sodium concentration under NC1 or 2DG treatment, cells were subjected to the previously described treatment durations and concentrations before being lifted off the culture plates.

Fluorescence microscopy

For fluorescence microscopy, HeLa cells transiently expressing the hTRPM4-EGFP fusion protein were seeded onto glass coverslips and grown to 70–80% confluency. Followed by appropriate NC1 treatment, the cells were loaded with 2.5 $\mu g\ ml^{-1}$ propidium iodide (PI) at room temperature for 20 min to stain the nuclei. After three washes with PBS, the coverslips were mounted onto glass slides and ready for fluorescence imaging.

For live-cell microscopy, cells were grown in four-well Lab-Tek II chambered coverglass (Thermo Fisher Scientific) imaging chambers. To image membrane potential changes, cells were incubated for 2 h with 5 μM DiBAC₄(3) (Dojindo) and then treated with 50 nM NC1 before imaging. To image the effect of ions on NECSO, MCF7 cells were incubated with SytoxGreen (Thermo Fisher Scientific) for 30 min at room temperature and the medium was replaced with another isotonic medium containing different ions as indicated before imaging.

TEM

TEM analyses were performed following established protocols. In brief, MCF7 cells were treated with 100 nM NC1 at different time points as

indicated and then fixed with a fixative solution containing 2% paraformaldehyde, 3% glutaraldehyde and 0.1 M cacodylate buffer at pH 7.3. Subsequently, the cells were washed with 0.1 M sodium cacodylate buffer and treated with 0.1% filtered cacodylate-buffered tannic acid. They were then postfixed with 1% buffered osmium and stained with 1% filtered uranyl acetate. After dehydration, the samples were subjected to a graded series of Epon up to 100% concentration for 6 h at 37 °C. They were then embedded in pure Epon and left overnight at 60 °C. Thin sections were obtained using a Leica EM UC7 ultramicrotome, placed on formvar-coated grids (Gilder Grids, AG100N) and stained with 3% uranyl acetate and lead citrate. Finally, the sections were examined using the TEM instrument PHILIPS CM-120 operating at 80 or 120 kV.

Protein expression and purification

hTRPM4 (NM_017636) containing a C-terminal 3 $\times$ Flag tag was cloned into a pCDNA4.1 vector and heterologously expressed in HEK293S cells (Life Technologies) using the PEI transfection system (Thermo Fisher Scientific). Cells were cultured in suspension at 37 °C for 48 h and harvested by centrifugation at 1,500 rpm for 15 min. The cell pellet was resuspended in buffer A (20 mM HEPES pH 7.4 and 150 mM NaCl) supplemented with a protease inhibitor cocktail (containing 2 $\mu g\ ml^{-1}$ DNase I, 0.5 $\mu g\ ml^{-1}$ pepstatin, 2 $\mu g\ ml^{-1}$ leupeptin, 1 $\mu g\ ml^{-1}$ aprotinin and 0.1 mM PMSF) and homogenized by sonication on ice. TRPM4 was extracted with 2% (w/v) DDM (Goldbio) supplemented with 0.2% (w/v) cholesteryl hemisuccinate (CHS; Sigma-Aldrich) by gentle agitation for 3 h at 4 °C. After extraction, the supernatant was collected following a 40-min centrifugation at 48,000g and incubated with Flag beads resin (Sigma-Aldrich) with gentle agitation overnight. Then, the resin was collected following a 5-min centrifugation at 300g and washed three times in buffer B (buffer A + 0.1% DDM + 0.02% CHS). TRPM4 was eluted with buffer B supplemented with 2 mg ml^{-1} Flag peptide and further purified by size-exclusion chromatography (SEC) on a Superose 6 10/300 GL column (GE Healthcare) pre-equilibrated with buffer B. The protein peak was collected and concentrated to use.

Protein–ligand interaction

For the CETSA, HEK293T cells stably expressing TRPM4 were cultured and treated with 100 μM NC1 or NC1i for 1 h at room temperature. Following compound treatment, cells were divided into aliquots and subjected to a heat shock at different temperatures ranging from 37 °C to 70 °C for 3 min. The heated samples were then immediately cooled on ice and lysed and the resulting lysates were centrifuged to obtain the soluble fraction. Protein levels in the soluble fraction were assessed with western blotting. The intensity of the protein bands was quantified using image analysis software (Fiji ImageJ, version 1.53a) and the thermal stability profiles were generated by plotting the protein band intensity against the heat-shock temperatures. A shift in the melting temperature (T_m) of the target protein in the presence of the compound indicates binding and stabilization of the protein. Control samples without compound treatment were included to establish baseline thermal stability profiles. Statistical analysis was conducted using GraphPad Prism 9 software and the significance of T_m shifts was determined using one-way analysis of variance (ANOVA) followed by Dunnett's post hoc test.

For the SPR-based binding assay, a Biacore 8K instrument (GE Healthcare) was used. The assays were conducted at 25 °C in HBS-EP running buffer (10 mM HEPES pH 7.4, 150 mM NaCl, 0.2% (v/v) surfactant DDM and 2% DMSO). The WT Htrpm4 protein was immobilized onto CM7 sensor chips using a standard amine-coupling procedure in 10 mM sodium acetate (pH 4.5). Compounds were diluted in a serial manner and injected onto the sensor chip at a flow rate of 30 $\mu l\ min^{-1}$ for 120 s (contact phase), followed by a buffer flow of 120 s (dissociation phase). The K_D value was determined using Biacore 8K evaluation

software (GE Healthcare, version 3.0.12) by analyzing the steady-state data at equilibrium.

Molecular docking simulations

The TRPM4 structure was obtained from the Protein Data Bank (PDB 6BQV), while the two-dimensional (2D) structure of NC1 or CLT was built from the ChemDraw software (CambridgeSoft, version 18.0). Molecular docking was performed using the Schrödinger Maestro software suite (Schrödinger, 2020-3). Before docking, the ligand was optimized using the OPLS3e force field in the LigPrep module. The protein underwent the protein preparation wizard, including the addition of missing residues, removal of waters and optimization of hydrogen bonds and energy using the OPLS3e force field. Protonation states of the protein and ligand at pH 7.4 ± 0.2 were determined using Epik. Docking was performed on a selected residue within a $25 \times 25 \times 25 \text{ \AA}^3$ grid box. Extra-precision docking (Glide XP) with flexible ligand sampling was used, followed by postdocking minimization to generate a maximum of ten ligand poses using the Glide program. The docking conformation with the highest score was analyzed to identify critical molecular interactions between the ligand and amino acid residues in the binding pocket, guiding site-directed mutagenesis.

MD simulation

For MD simulation, only the TM domain of TRPM4 was selected. The best pose of the TRPM4–NC1 complex obtained from docking was chosen and the ligand–protein systems were constructed using the CHARMM-GUI⁶² (<https://charmm-gui.org>) membrane builder. The complex was inserted into a lipid bilayer composed of 180 1-palmitoyl-2-oleoyl-*sn*-glycero-3-phosphocholine molecules, solvated with a 22.5-Å-thick layer of 150 mM NaCl aqueous solution. To assess the stability of the ligand–protein poses, 50-ns simulations were performed. After 25,000 steps of minimization, the systems underwent equilibration in six steps according to the default settings from CHARMM-GUI lipid bilayer outputs, followed by a 50-ns production run. All simulations were conducted using GROMACS 2020.3 with CHARMM36m force fields and WYF parameters for the protein, CHARMM general force field for the ligand, CHARMM36 force fields for the lipids and the TIP3P model for water. The simulations used isothermal–isobaric (NPT) ensembles at a constant temperature of 303.15 K. The simulation trajectories were analyzed for structural stability using r.m.s.d. calculations. MD simulation trajectories were also subjected to molecular mechanics energies combined with Poisson–Boltzmann surface area continuum solvation calculations using a `gmx_mmpbsa` bash script⁶³, with solvent-accessible surface area used as the model for non-polar solvation energy. The best pose of the TRPM4–NC1 complex from molecular docking was selected for binding pocket analysis using Fpocket 2.0 software. Pocket parameter information was generated using the `dpocket` program with default settings.

Statistics and reproducibility

Statistical analyses, including one-way and two-way ANOVA, were conducted using GraphPad Prism software (version 9.0.0). All statistical data are reported as the mean $\pm$ s.d., except for the patch-clamp data, which are presented as the mean $\pm$ s.e.m. All microscopic, biochemical and biological assays were independently repeated at least three times with similar results. The mean values of the control and experimental groups were compared for significant differences. Data distribution was assumed to be normal, although formal testing was not performed. *P* values below 0.05 were considered statistically significant. The sample size was not determined using a statistical method. No data were excluded from the analyses. Illustrations were prepared using Adobe Illustrator and BioRender.com. The experiments were not randomized and the investigators were not blinded to treatment allocation or outcome assessment, unless otherwise specified.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All original data that support the findings of this study are provided in the article and Supplementary Information. All specific and stable reagents generated in this study are available from the corresponding authors without restriction. Source data are provided with this paper.

References

- Huang, J. et al. Analysis of ionic profiles of spinal cords in a rat model with bone cancer pain. *J. Pain Res.* **17**, 1531–1545 (2024).
- Jo, S., Lim, J. B., Klauda, J. B. & Im, W. CHARMM-GUI Membrane Builder for mixed bilayers and its application to yeast membranes. *Biophys. J.* **97**, 50–58 (2009).
- Valdes-Tresanco, M. S., Valdes-Tresanco, M. E., Valiente, P. A. & Moreno, E. `gmx_MMPBSA`: a new tool to perform end-state free energy calculations with GROMACS. *J. Chem. Theory Comput.* **17**, 6281–6291 (2021).

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Author contributions

W.F. and J.W. performed most of the biological and biochemical experiments. T.L. performed electrophysiological experiments with the help of R.X., Y.H. and Z.L. J.W. performed the molecular docking, MD simulations and mutagenesis analysis. Y.Q. and Z. Zhang performed the chemical inhibitor screening and helped with some of the cell experiments. X.Z., M.H., Y.S. and Z. Zhao helped with the cell biological experiments. C.L. performed the CRISPR screening data analysis with guidance of J.L. S.Z. and G.C. helped in data analysis. W.F. and Q.Z. conceptualized the project and designed the experiments. W.F., J.W. and Q.Z. analyzed the data and wrote the paper with the help of all authors. Y.L. and Q.Z. supervised the project. All authors discussed the results and commented on the paper.

Competing interests

The authors declare no competing interests.

Additional information

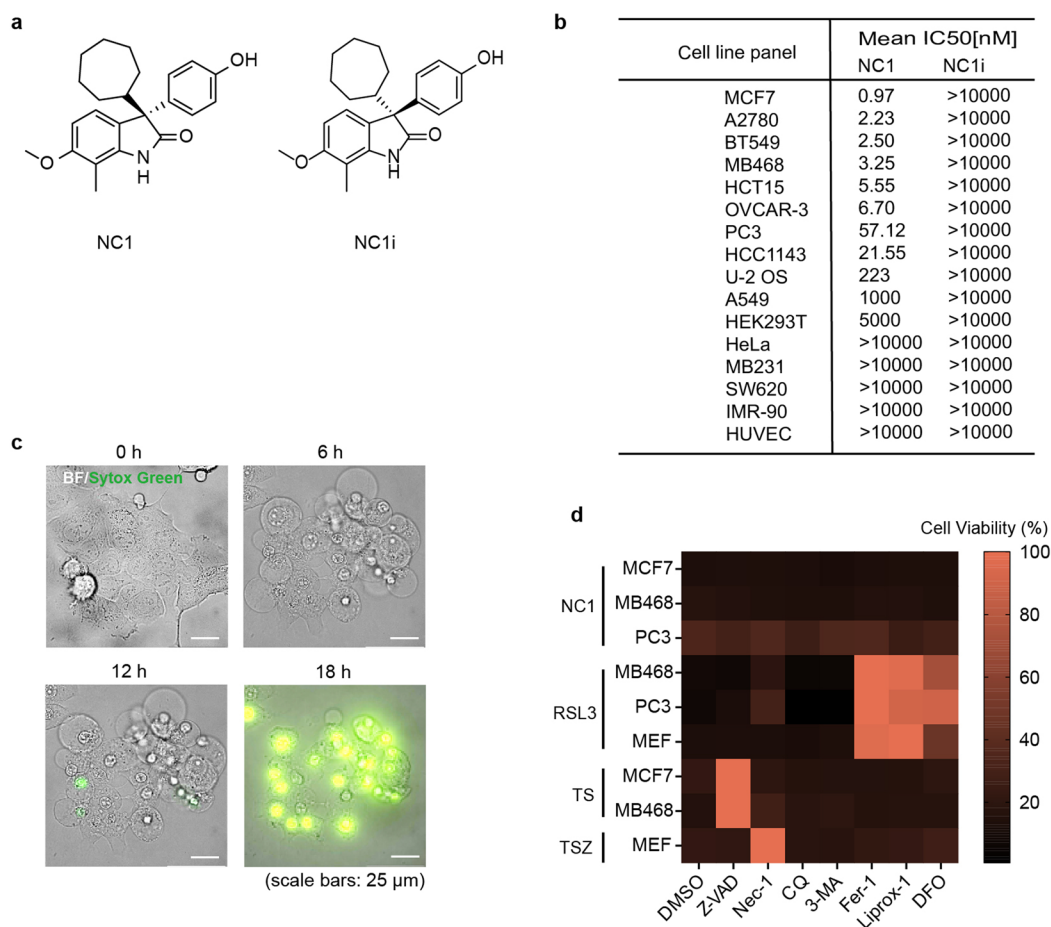
Extended data is available for this paper at <https://doi.org/10.1038/s41589-025-01841-3>.

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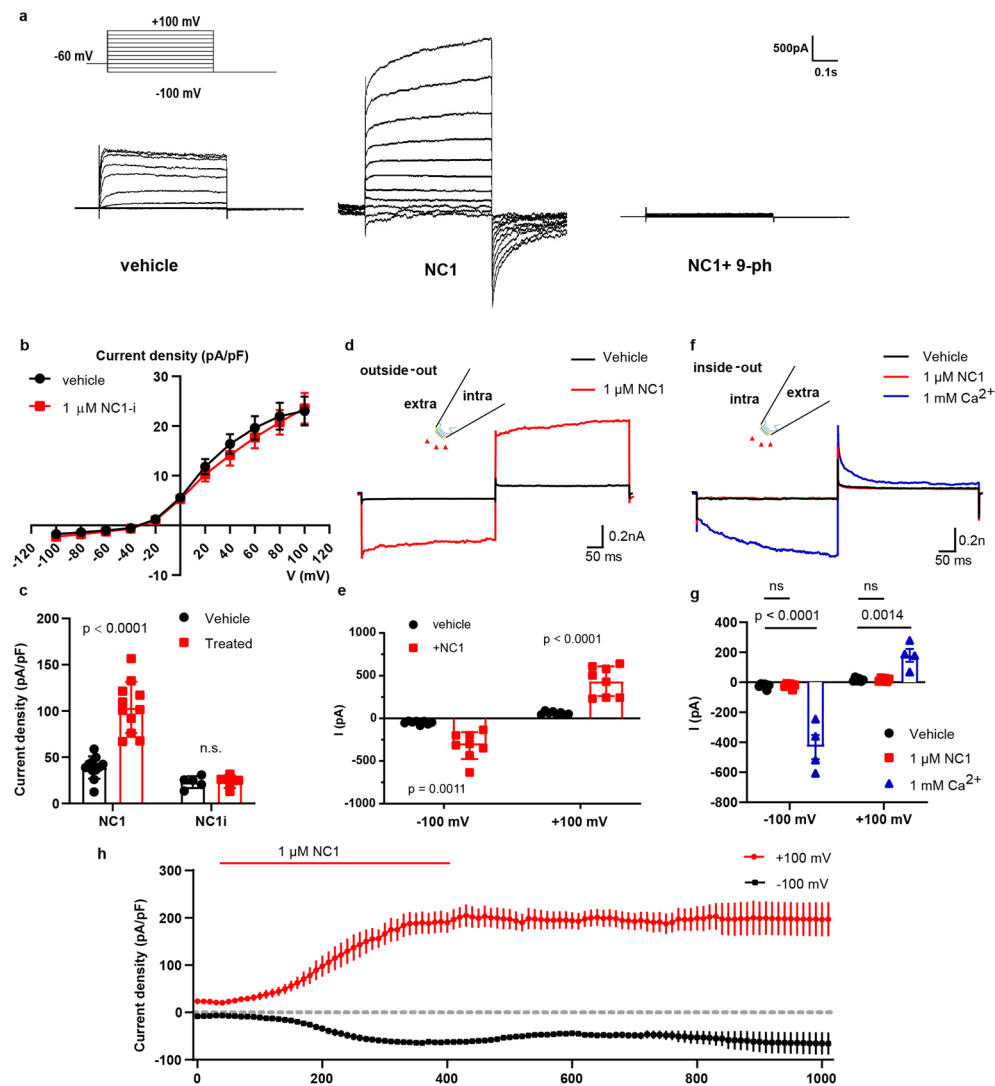
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Extended Data Fig. 1 | Different death-inducing effect of the stereoisomers NC1 and NC1i. **a**, Chemical structure of compound NC1 and NC1i. **b**, IC₅₀ of NC1 and NC1i in various cell lines treated for 48 h. **c**, Live-cell images of MCF7 cells treated with 50 nM NC1 in medium containing SytoxGreen. Scale bars, 25 μm. **d**, Heatmap of viability of cells pretreated for 1 h with 20 μM Z-VAD-FMK (Z-VAD), 10 μM Necrostatin-1 (Nec-1), 50 μM Chloroquine (CQ), 50 μM 3-Methyladenine

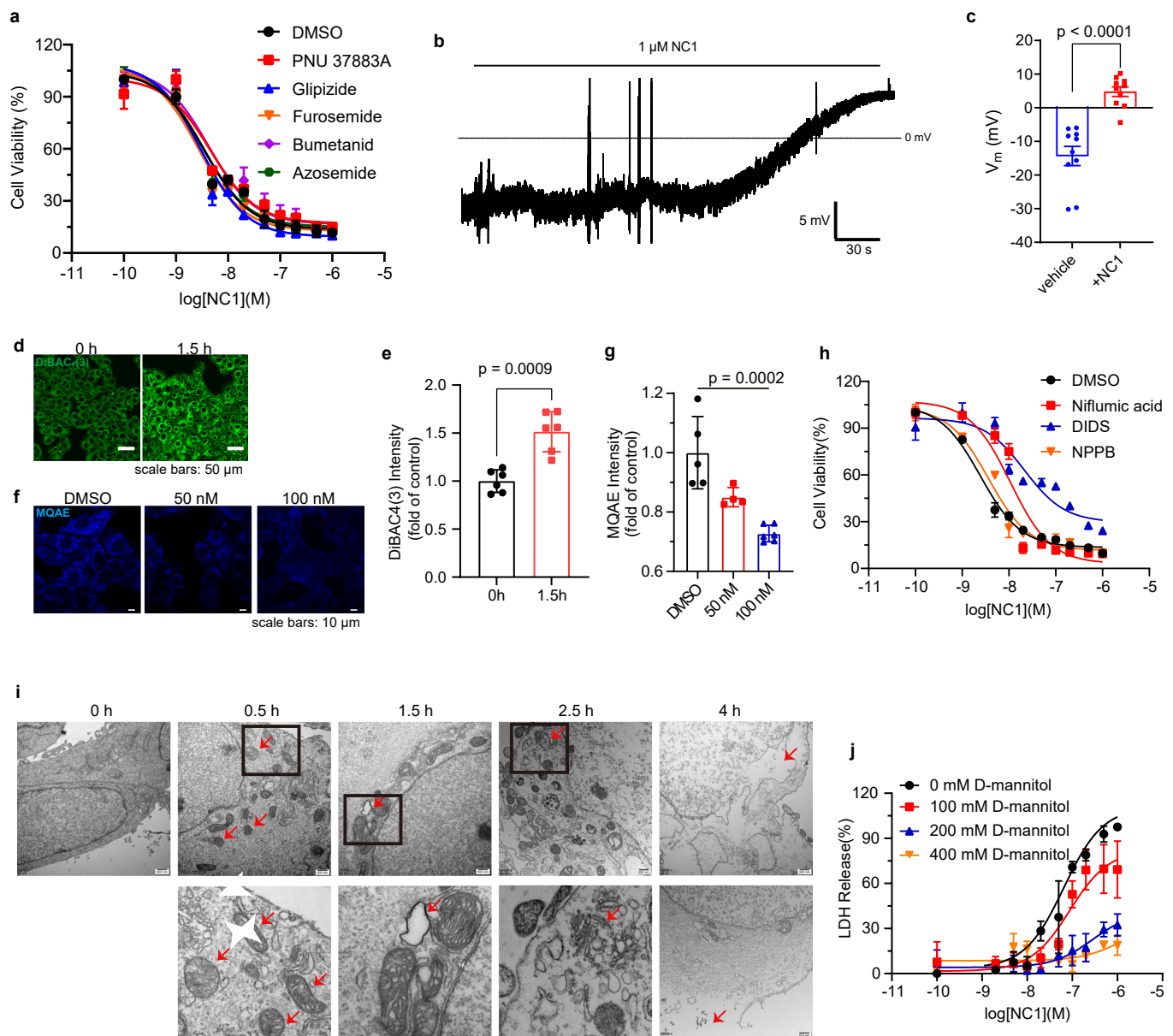
(3-MA), 1 μM Ferostatin-1 (Fer-1), 1 μM Liproxstatin-1 (Liprox-1), or 100 μM Deferoxamine (DFO) and then treated with either 100 nM NC1, RSL3, 50 ng/ml TNF-α + 1 μM Smac Mimetic (TS), or TS + 20 μM Z-VAD (TSZ) for 24 h. Viability data represent mean of *n* = 3 replicates from one representative of three independent experiments. Micrograph data show one representative out of three replicates from three independent experiments.



Extended Data Fig. 2 | NC1, rather than NC1i, triggers TRPM4 activation.

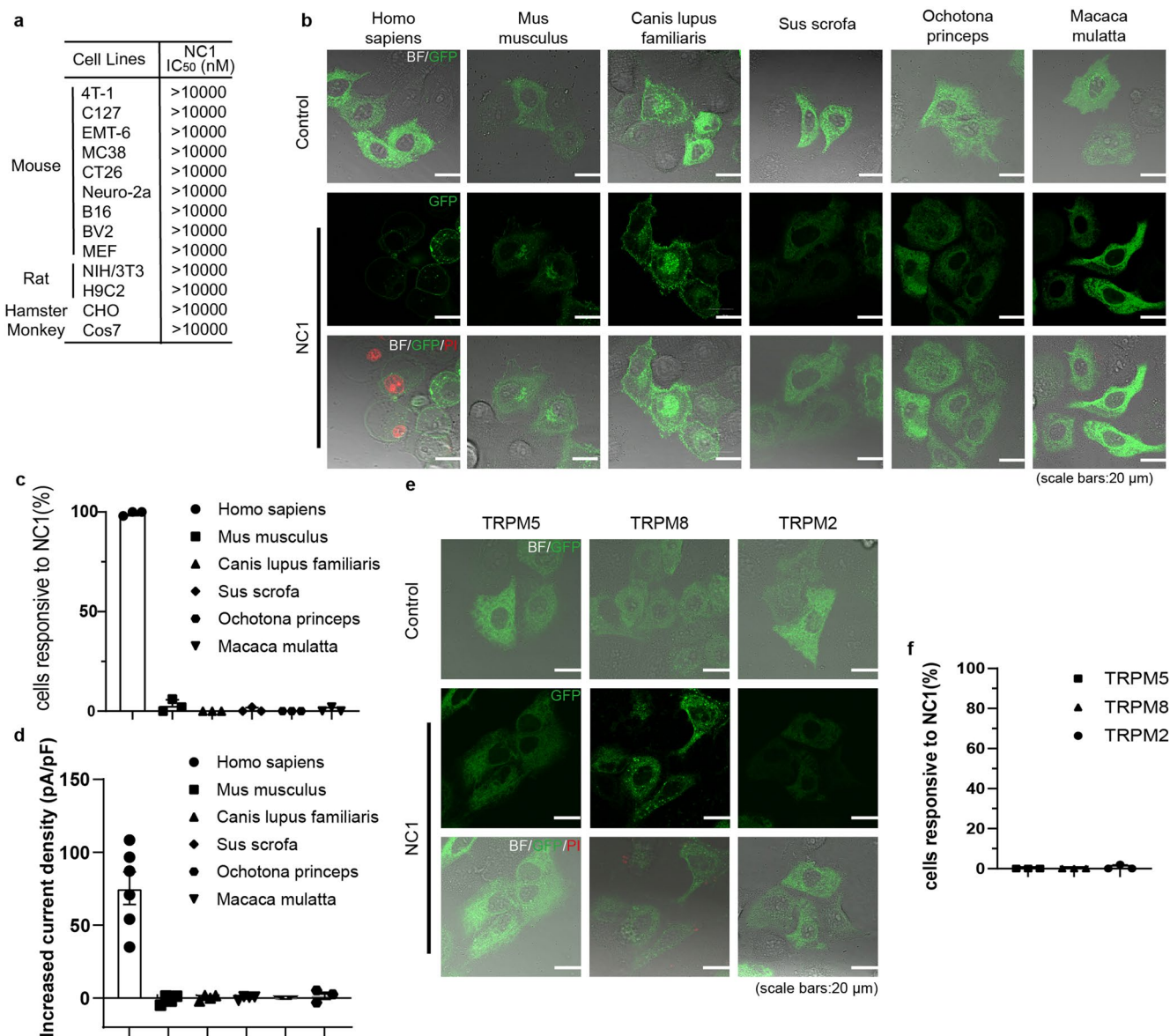
a, Typical whole-cell current traces recorded from HEK293T cells stably expressing hTRPM4 channel and stimulating voltage steps used in the experiment is on the top (left panel). Subsequently, whole-cell current traces recorded after 1 μ M NC1 application (middle panel), followed by inhibition with 50 μ M 9-ph inhibitor (right panel). Three figures were recorded from one same patch. **b**, The current density-voltage relation of hTRPM4 steady-state current with vehicle or NC1i application. **c**, Statistics of hTRPM4 steady-state whole-cell current density with vehicle, NC1i or NC1 application at +100 mV in **b**. **d-g**, Typical

current traces of hTRPM4 channel treated vehicle or 1 μ M NC1 by using outside-out (**d**) or inside-out (**f**) patch. Statistics of hTRPM4 outside-out (**e**) or inside-out (**g**) current. **h**, The time course of TRPM4 current-densities in HEK293T-TRPM4 cells activated by 1 μ M NC1 was followed by NC1 washout using bath solution after 420 seconds. The whole-cell currents were recorded at -100 and +100 mV. Electrophysiological data represent mean $\pm$ sem of (n = 5 for **b**, 11 for NC1 and 5 for NC1i in **c**, 8 for **e**, 4 for **g**, 8 for **h**) independent cells. Two-way ANOVA with post Sidak test (GraphPad Prism 9.0.0).



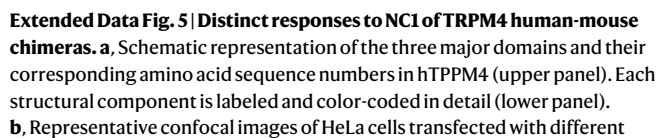
Extended Data Fig. 3 | NECSO is accompanied with Membrane depolarization and cell edema. **a**, Dose-dependent cytotoxicity of NC1 in MCF7 cells pretreated with 20 μ M NKCC inhibitors (Furosemide, Bumetanid and Azosemide) or KATP inhibitors (PNU 37883A and Glipizide). **b**, A typical membrane potential of MCF7 cells treated with 1 μ M NC1. **c**, Statistics of membrane potential values of MCF7 cells recorded 10 minutes after exposed to vehicle or NC1 in **b**. **d**, Representative confocal images of MCF7 cells loaded with 5 μ M DiBAC4(3) followed by treatment with 50 nM NC1 for 1.5 h. **e**, Statistics of fold change for DiBAC4(3) fluorescence intensity, mean $\pm$ s.d. Each dot represents an individual image in **d**. **f**, Representative confocal images of MCF7 cells treated with DMSO, 50 or 100 nM NC1 for 3 h and then loaded with chloride indicator MQAE. Scale bars, 10 μ m. **g**, Statistics of fold change MQAE fluorescence intensity, mean $\pm$ s.d. Each dot represents an individual image in **f**. **h**, Dose-dependent cytotoxicity of NC1

in MCF7 cells pretreated with multiple inhibitors of chloride anion channels. **i**, Transmission electron microscopy (TEM) of MCF7 cells treated with 100 nM NC1 at different time points as indicated. Scale bars, 500 nm. Red arrowheads in images of 0.5, 1.5 and 2.5 h represented swollen mitochondria, endoplasmic reticulum and Golgi apparatus respectively, and highlighted swollen nuclear membrane (above) and broken cell membrane (below) in images of 4 h. **j**, Cell death assay measured by LDH release of MCF7 cells treated with NC1 for 24 h in medium containing different doses of osmo-protectant D-mannitol. Viability and cell death data represent mean $\pm$ s.d. of $n = 3$ replicates from one representative of three independent experiments. Electrophysiological data represent mean $\pm$ sem of $n = 10$ independent cells. Micrograph data show one representative out of three replicates from three independent experiments. One-way ANOVA for statistics in **c**, **e**, **g** (GraphPad Prism 9.0.0).

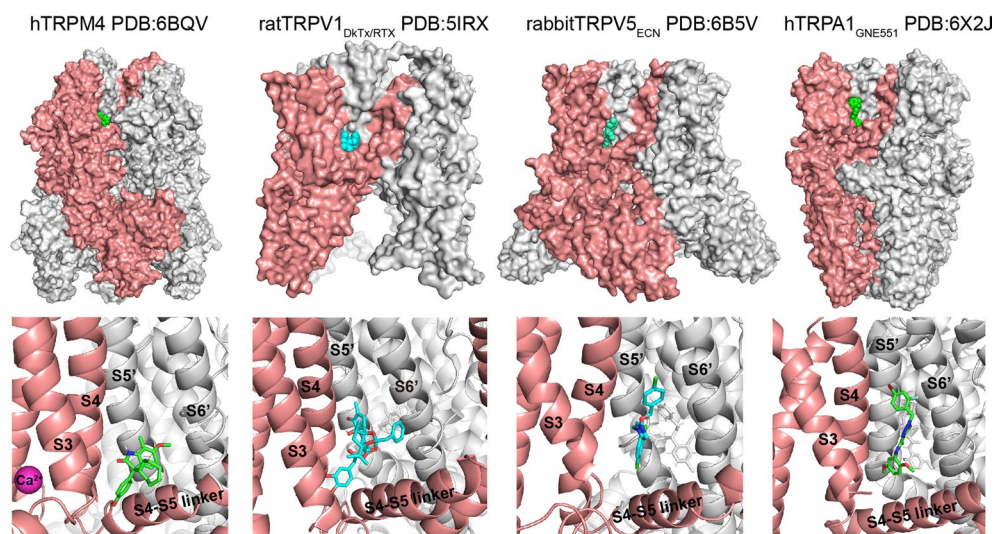


Extended Data Fig. 4 | Different responses to NC1 of TRPM4 from multiple species. **a**, IC₅₀ of NC1 in various cell lines of different species treated for 48 h. **b**, Representative confocal images of HeLa cells with expression of various species TRPM4, treated with 1 μ M NC1 for 24 h. PI staining indicates the dead cell. **c**, Statistics of the percentage of NC1-responsive cells in TRPM4-EGFP transfected cells in **b**, mean $\pm$ s.d. from 50 cells from three independent experiments. **d**, Summary results of sensitivity to NC1-induced current density

of different species TRPM4 at +100 mV in whole-cell patch clamp experiments. **e**, Representative confocal images of HeLa cells transfected with different TRPM family protein, subsequently treated with 1 μ M NC1 for 24 h. PI staining indicates the dead cell. **f**, Histogram summarizes the percentage of NC1-responsive cells in TRPM4-EGFP transfected cells in **e**, mean $\pm$ s.d. from 50 cells from three independent experiments. Micrograph data show one representative out of three replicates from three independent experiments.

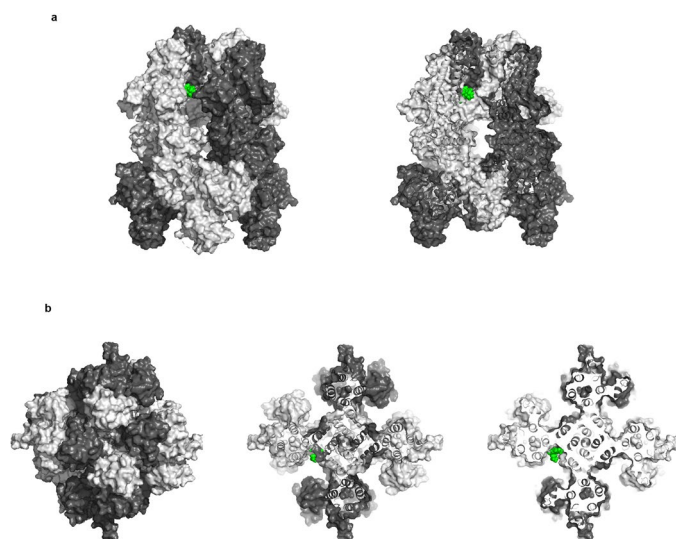


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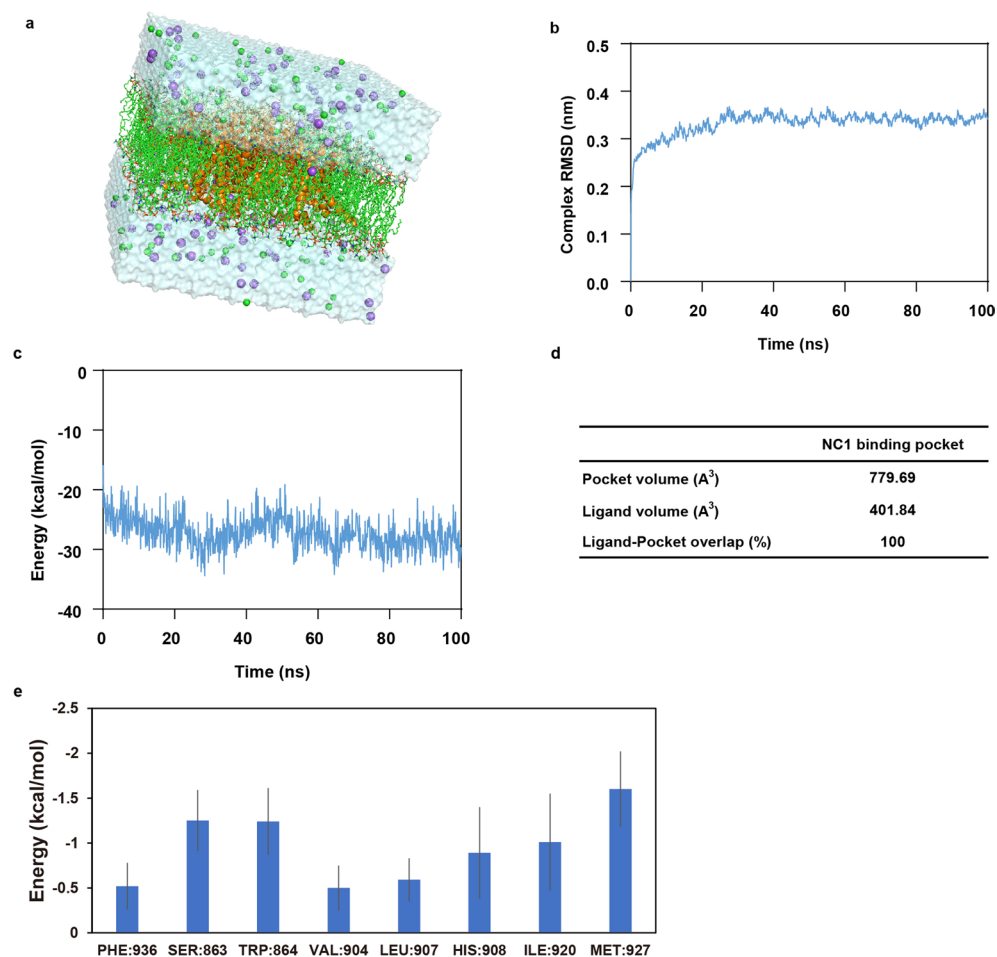


Extended Data Fig. 6 | Model representations of ligand binding sites in vanilloid binding pocket (VBP). The overall architecture of the indicated TRP channels, and one subunit is highlighted. Ligands are presented with spheres

(upper panel). Ligands are shown as sticks. Oxygen is shown in red, nitrogen in blue, sulfur in yellow, fluorine in light blue, bromine in dark red, and chloride in green (lower panel).

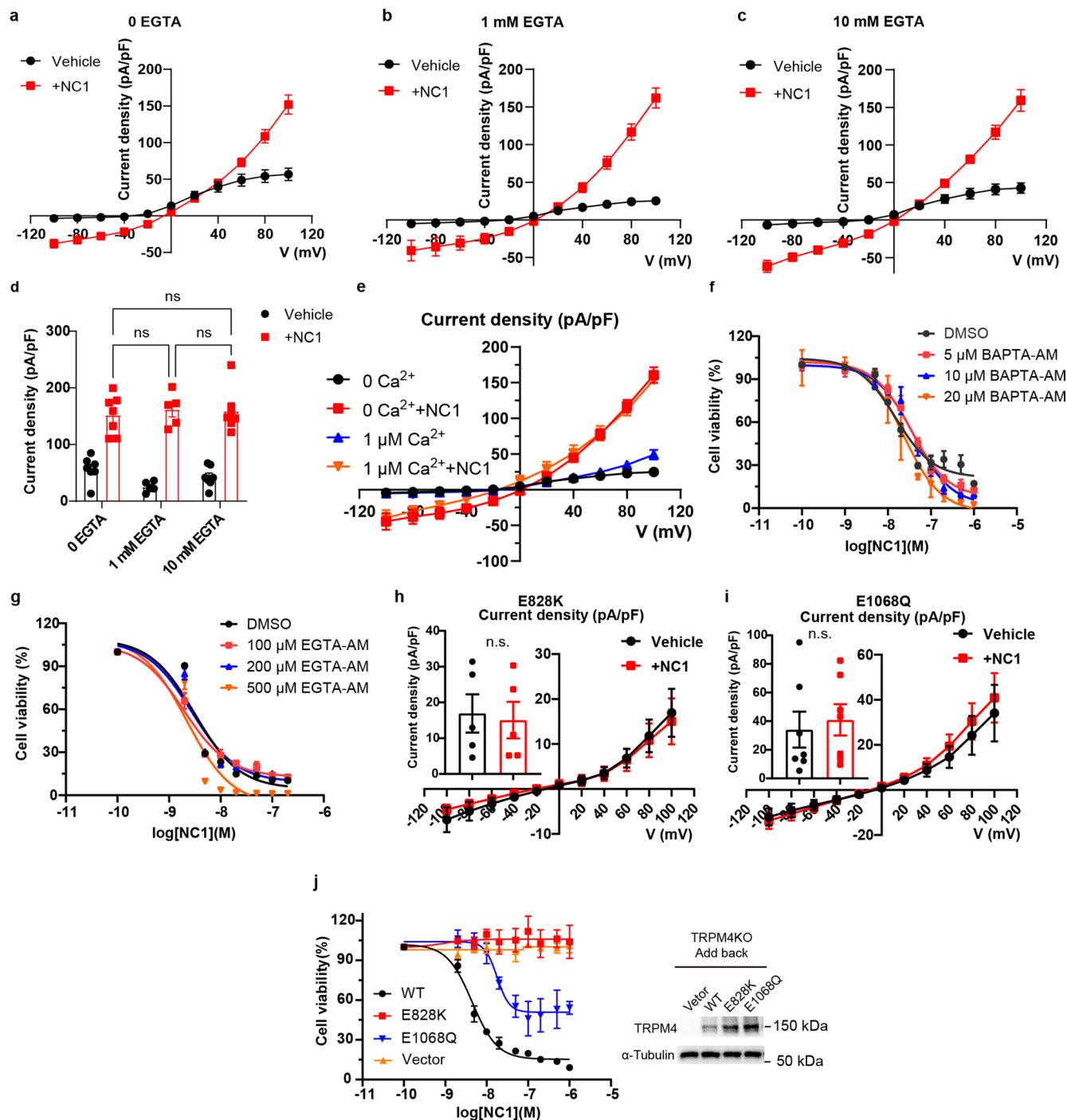


Extended Data Fig. 7 | Different cross-sections of Ligand NCl binding pocket in hTRPM4 (PDB: 6BQV). **a, b,** Overall surface presentation of TRPM4. Ligand NCl is shown in green spheres presentation. Subunits are individually colored with gray black or gray white. The ligand binding pocket of NCl is composed of residues from two adjacent subunits. The view is from the front (a) or the top (b).



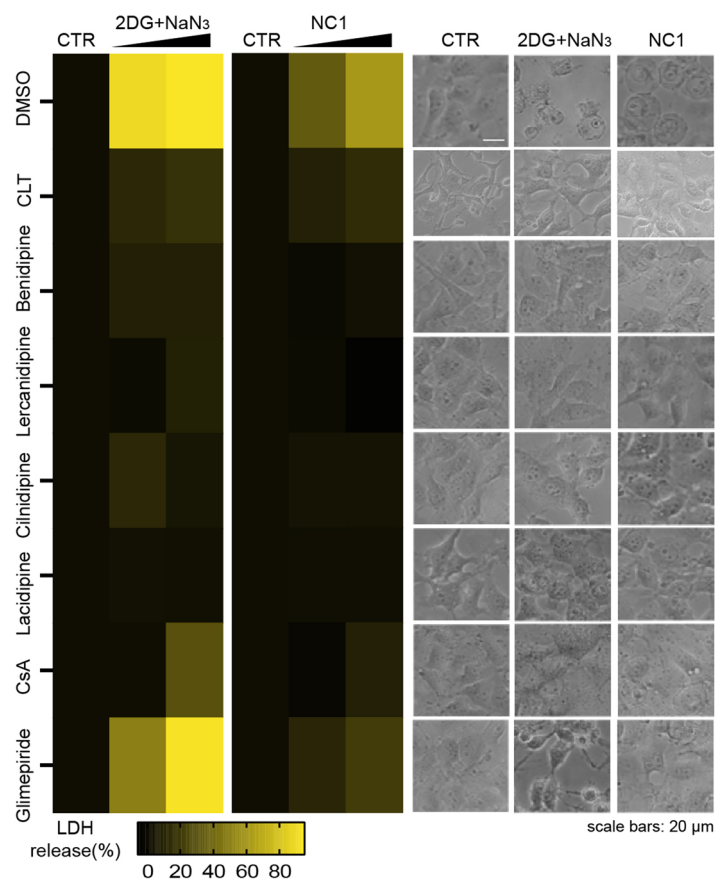
Extended Data Fig. 8 | Molecular dynamics (MD) simulation analyses of the binding of NC1 to TRPM4. **a**, The simulated state diagram of the TRPM4-NC1 complex embedded in the cell membrane phospholipid bilayer. **b**, Summary results of structural stability analyses (root mean square deviation, RMSD) of channel-ligand complex following NC1 binding to TRPM4 channel in 100 ns bouts. **c**, Interaction energy calculations for NC1 binding to TRPM4 channel

on trajectories with convergent conformations. **d**, Information on the binding pockets of NC1 with the TRPM4 channel was calculated after MD simulations. **e**, Energy contribution of residues within a 5 Å distance of NC1. The mean values of energy are represented by blue bars, while the black lines depict the standard error.



Extended Data Fig. 9 | NC1 activation of TRPM4 per se is not affected by calcium. **a, b, c**, The conductance-voltage relation of whole-cell hTRPM4 current in HEK293T-TRPM4-EGFP cells treated vehicle or 1 μM NC1 with different concentrations of EGTA (0, 1, 10 mM) added in the patch pipette solutions. **d**, Statistics of whole-cell hTRPM4 current density at +100 mV in a, b, c. **e**, The current density-voltage relation of whole-cell hTRPM4 current in HEK293T-TRPM4 cells treated vehicle or 1 μM NC1 in the presence or absence of 1 μM intracellular Ca^{2+} . NC1 further increase the current amplitude of TRPM4 with the presence of intracellular calcium. **f, g**, Dose-dependent cytotoxicity of NC1 in MCF7 cells pretreated with calcium ion chelators BAPTA-AM (f) or

EGTA-AM (g). **h, i**, The whole-cell current-voltage relation of hTRPM4 E828K (h) or E1068Q (i) mutant expressed in HeLa cells (right panel). Summary data showing the maximum amplitude of currents activated by NC1 at +100 mV (left panel). **j**, Dose-dependent cytotoxicity of NC1 in TRPM4-KO MCF7 cells with re-expression of TRPM4 WT or mutants above. Viability data represent mean $\pm$ s.d. of $n = 3$ replicates from one representative of three independent experiments. Electrophysiological data represent mean $\pm$ sem of ($n = 7$ for a, e and i, 5 for b and h, 6 for c) independent cells. Immunoblot data show one representative out of three replicates from three independent experiments. Two-way ANOVA (GraphPad Prism 9.0.0) for statistics in d, h, i.



Extended Data Fig. 10 | NEC50 inhibitors protect cells from energy depletion. Heatmap depicting the cell death (measured by LDH release) of Cos7 cells stably expressing hTRPM4 which were pre-treated with different inhibitors followed by 2DG + NaN₃ or NC1 exposure respectively for 16 h (left). Representative

bright field images of the cellular morphology corresponding to each treatment condition as described above (right). Scale bars, 20 μm. Cell death and micrograph data show one representative out of three replicates from three independent experiments.

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
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☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☒	☐ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
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☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	We used commercial software available with the respective instruments for data collection. Absorbance-based assay data were collected on an VICTOR Nivo fluorometer(Perkin Elmer). Westernblot data was collected on an Odyssey machine and Image Studio software-Ver 5.2 (LiCor) or Bio-Rad ChemiDoc Imaging System. Flow cytometry data was collected on CytoFlex equipment and CytoExpert software-Ver 2.3 (Beckman Coulter. Fluorescence images were taken by the FV3000 confocal microscope(Olympus Corporation), or IX83 microscope(Olympus Corporation), other data collection methods were specified in the methods section describing each experimental method.
Data analysis	All software used to process and plot data in this study were listed in the methods section. Graphs and numerical data (including statistics/ error bars)were analyzed and plotted by Prism(GraphPad, Version 9.0.0); Image J(NIH, Version 1.53a)for the analysis of microscopy data; Flow cytometry data was analyzed using FlowJo 10. The gRNA sequencing data were calculated and analyzed using the MAGeCKFlute with R(v 4.0.5). 2D structure of NC1 or CLT was built from the ChemDraw software (CambridgeSoft, v18.0). Molecular docking was performed using the Schrödinger Maestro software suite (Schrödinger, 2020-3). All molecular dynamics simulations were conducted using GROMACS (v2020.3). Other data analysis methods were specified in the methods section describing each experimental method.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

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All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The TRPM4 structure was obtained from the Protein Data Bank (PDB ID: 6BQV). All original data that support the findings of this study are provided in source data and supplementary information.

Human research participants

Policy information about [studies involving human research participants and Sex and Gender in Research](#).

Reporting on sex and gender

N/A

Population characteristics

N/A

Recruitment

N/A

Ethics oversight

N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

Sample sizes were determined according to the related references (PMID: 22632970, PMID: 35298263, PMID: 37015922, PMID: 29211723) and chosen based on logistical and experimental experiences. In every case, three or two biological replicates for each experiment were included. All experiments were reliably reproduced and the number of samples chosen was sufficient to support meaningful conclusions.

Data exclusions

In general, no data were excluded, except for experimental failures due to control failures and technical problems.

Replication

All microscopic, biochemical, and biological assays were independently repeated at least three times as mentioned in Methods or figure legends. All attempts at replication were successful.

Randomization

All groups were randomly assigned, with each group representing a different treatment or condition.

Blinding

Blinding was not applicable to this study as the data grouping and analysis were performed in an unbiased manner. Data was analyzed with common strategies.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	Anti-TRPM4 Antibody (Abclonal, Cat # A10146, diluted 1:1000) Anti-α-Tubulin Antibody (Sigma Aldrich, Cat # T6199, diluted 1:10000) Anti-β-actin Antibody (Sigma Aldrich, Cat # A5441, diluted 1:10000) Anti-Cas9 Antibody (Abcam, Cat # ab204448, diluted 1:1000) Peroxidase-conjugated AffiniPure Goat Anti-Rabbit IgG secondary Antibody (Jackson ImmunoResearch Laboratories , Cat # 111-035-003, diluted 1:10000) Peroxidase-conjugated AffiniPure Goat Anti-Mouse IgG secondary Antibody (Jackson ImmunoResearch Laboratories , Cat # 115-035-003, diluted 1:10000)
Validation	All information and validation of commercial antibodies used in this work are available on the manufacturers' websites listed below: Anti-TRPM4 Antibody (Abclonal, Cat # A10146, diluted 1:1000) https://abclonal.com.cn/catalog/A10146 Anti-α-Tubulin Antibody (Sigma Aldrich, Cat # T6199, diluted 1:10000) https://www.sigmaaldrich.cn/CN/zh/product/sigma/t6199 Anti-β-actin Antibody (Sigma Aldrich, Cat # A5441, diluted 1:10000) https://www.sigmaaldrich.cn/CN/zh/product/sigma/a5441 Anti-Cas9 Antibody (Abcam, Cat # ab204448, diluted 1:1000) https://www.abcam.cn/products/primary-antibodies/crispr-cas9-antibody-ab204448.html Peroxidase-conjugated AffiniPure Goat Anti-Rabbit IgG secondary Antibody (Jackson ImmunoResearch Laboratories , Cat Num #111-035-003) https://www.jacksonimmuno.com/catalog/products/111-035-003 Peroxidase-conjugated AffiniPure Goat Anti-Mouse IgG secondary Antibody (Jackson ImmunoResearch Laboratories , Cat Num #115-035-003); https://www.jacksonimmuno.com/catalog/products/115-035-003

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)	MCF7, MB468, HeLa, HEK293T, SW620, U-2 OS, COS-7, A549, PC3, HCC1143, A2780, HCT15, OVCAR-3, IMR-90, HUVEC, CHO, AC16, B16, MC38 and HEK293S cells were obtained from ATCC. BT549, MDA-MB-231, H9C2, NIH/3T3, C127, 4T-1, EMT-6, CT26.WT, BV2, Neuro-2a cells were purchased from Procell. MCF7-Cas9, MCF7-TRPM4KO, MB468-TRPM4KO, AC16-TRPM4KO, HEK293T-hTRPM4-EGFP, HEK293T-mTRPM4-EGFP, COS-7-hTRPM4-EGFP and MEF cells were generated in this study.
Authentication	No further authentication of the cell lines was performed before use.
Mycoplasma contamination	All cell lines used in experiments were tested negative for mycoplasma contamination.
Commonly misidentified lines (See ICLAC register)	No commonly misidentified cell lines were used in our study.

Plots

Confirm that:

- ☐ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☒ All plots are contour plots with outliers or pseudocolor plots.
- ☐ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	MCF7 cells were detached from culture plates using 1X PBS + EDTA. Single cell suspensions were prepared in 1X PBS with 4% FBS and then stained with 2 μ M CoroNa-AM dye at 37°C for 30 minutes, followed by two washes with 1X PBS 4% FBS.
Instrument	CytoFlex S instrument (Beckman Coulter)
Software	FlowJo10
Cell population abundance	A minimum of 50,000 cells were analyzed per condition.
Gating strategy	FSC-A vs SSC-A gating was used to identify cells of interest based on size and granularity. FSC-A vs FSC-H density plot was used to exclude doublets. Cells within the gate were analyzed for CoroNa probe using FITC filter.
☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.	